ETHEKWINI EMERGENCY SERVICES

Criticom V5 Operations & Engineering Manual

How to run the platform and how it is built — every role, every task, every screen, step by step.

I The Platform · **II** Technology · **III** Operational Flow
IV Role Playbooks · **V** Task Procedures · **VI** The Responder App
VII Administration · **VIII** Running It · **IX** Reference

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Contents

VOLUME I — THE PLATFORM

1 What Criticom is · the problem · the five departments · lineage · in numbers

2 Architecture · topology · logic and authorization in the database · repository layout

3 Authorization · silos · roles · route matrix · row-level security · privileged functions

VOLUME II — TECHNOLOGY

4 The stack · every dependency, its version and why it is there

5 How a request works · render path · mutation path · realtime path · offline path

VOLUME III — OPERATIONAL FLOW

6 The life of a call · eleven steps, screen by screen

7 The status model · every status, who may set it, what it means

8 The alert model · severity, presentation, acknowledgement, delivery

VOLUME IV — ROLE PLAYBOOKS

9 How to read a playbook

10 Call-taker · IncidentLogger

11 Control room operator · ControlRoom, Operator

12 Dispatcher · Dispatcher

13 Field responder · Responder

14 Metro Police officer · Police, Officer, Metro, Security, Captain

15 Firefighter · Firefighter, Chief, FireChief

16 Paramedic · Paramedic, Doctor

17 Disaster management officer · Disaster

18 Supervisor · Supervisor, Captain, Chief, FireChief

19 Case manager · CaseManager

20 Staff and Viewer · Staff, Viewer

21 Administrator · Admin, SystemAdmin, SuperAdmin

VOLUME V — TASK PROCEDURES

22 Intake and dispatch · log a call · route · assign · duplicate · citizen GPS

23 Working an incident · status · updates · evidence · close · PDF

24 Start of shift · mental check · vehicle selection · inspection

25 Departmental reports · police · fire · PCR · security · firearms

26 People · roster · allocation · leave · training · certifications · performance

27 Reporting and analysis · report hub · builder · KPI exports · hierarchy

28 Surveillance and external systems · cameras · footage · PTT · feeds

VOLUME VI — THE RESPONDER APP

29 Screens and the duty gate

30 Mobile task procedures · every action on the phone

31 The offline outbox

32 Officer-down

33 Alerting and push

VOLUME VII — ADMINISTRATION

34 Users, roles and personnel

35 Configuration · categories · leave types · shift types · ops links

36 Operations settings · all twelve knobs

37 Automations · the seven rules and how to tune them

38 Integrations · catalogue · proxy pattern · CCTV cut-over

39 Branding

VOLUME VIII — RUNNING IT

40 Security · model · headers · rate limits · audit · outstanding items

41 Deploying · web · database · mobile · the OTA trap

42 Backup and recovery

43 First response · triage order · common situations

44 Engineering standards · CI · testing · linting · conventions

VOLUME IX — REFERENCE

A Action index · all 67 server actions and their fields

B Route index · **C** API endpoints

D Data model · **E** Stored procedures · **F** Migrations

G Environment variables · **H** Glossary

VOLUME I

The Platform

What Criticom is for, how it is put together, and the authorization model that makes one shared incident table safe for five separate emergency services.

CHAPTER 1

What Criticom Is

> One call-taker logs an incident; the platform routes it to the right departments, puts it on a map, pushes it to the crews' phones, records everything they do at the scene, and keeps the evidential trail that an inquiry will read a year later.

1.1 The problem it solves

Five emergency services operate across eThekwini. They share a city, they share incidents, and they routinely need each other — a structure fire needs Fire and EMS and often Police for the cordon. What they did not share was a system: each kept its own records, so a call touching three services produced three partial accounts, joined only by the radio.

Criticom's premise is that the incident is the record and every department writes to the same one. Three properties follow:

- **Departments see their own work and no more** — but when a call is routed to several, all of them see it. Resolved once, in the database.
- **The record is evidence.** It gets read in disciplinary hearings, insurance disputes and inquests. A timestamp recording "when the network came back" rather than "when the crew arrived" is worse than none, because it looks authoritative.
- **Nothing may depend on good conditions.** Coverage fails, phones lock, screens get ignored during a major incident. Each of those degrades rather than loses data.

1.2 The five departments

Departments are **silos** in the code and database. The term names the isolation boundary, which is a security concept.

SILO	SERVICE	SCOPE
`Ethekwini`	Control Room	Global. Takes calls, routes them, sees every department's incidents.
`MetroPolice`	Metro Police	Law enforcement, by-law, traffic, firearms register, SIRA guard reports.
`MetroFire`	Fire & Rescue	Suppression, rescue, hydrant register, post-incident review.
`Medical`	EMS	Ambulance response, patient care records, hospital handover, restocking.
`EthekwiniDisaster`	Disaster Management	Floods, major incidents, mutual aid, relief stock, post-mortem.

1.3 Lineage from V4

A rebuild of an ASP.NET MVC application of ~38 controllers over five SQL Server databases. V5 reproduces every capability and consolidates into one Postgres schema, with row-level security doing the separation the separate databases used to do badly. Five waves: 1–3 reached parity, 4 aligned a July feature drop, 5 added offline field operation, officer-down, automatic escalation, search, rate limiting and tests.

1.4 In numbers

110	67	57	41	25
WEB ROUTES	USER ACTIONS	TABLES	PROCEDURES	ROLES

28
MIGRATIONS

CHAPTER 2

Architecture

2.1 Topology

COMPONENT	RUNS ON	RESPONSIBILITY
Console	Vercel · Next.js 15, React 19	Every operator screen. Server-rendered per request.
Responder app	Expo SDK 52 · React Native 0.76	The crew's phone. EAS distribution; JS ships over the air.
Database	Supabase Postgres 15 + PostGIS	Data, authorization, realtime, storage, and much of the logic.
Automations	`pg_cron`, in-database	Escalation, expiry, digests. No worker to deploy.

2.2 Logic in the database

Dispatching writes an incident row, updates unit state, appends a timeline entry, writes an audit record and raises one alert per receiving department. As application code that is six round trips that can half-succeed — and a half-dispatched call is a crew that never rolls. So those workflows are stored procedures, and web and mobile call the same one, so the two surfaces cannot drift.

> **THE TRADE-OFF**
> SQL is harder to test and harder for a TypeScript developer to change. It is accepted only where atomicity is the point. Reporting, KPI arithmetic and formatting stay in TypeScript, where they are readable and unit-tested.

2.3 Authorization in the database

Row-level security decides what each user reads and writes. The application never adds a `WHERE silo = …` for security — only to cut traffic. This is the most important decision in the system, and it is what makes a shared incidents table safe for five services. A bug in a page cannot leak another department's data, because the page never had it.

2.4 Repository layout

PATH	CONTENTS
`apps/web`	The console. Route groups `(app)` authenticated, `(auth)` login.
`apps/mobile`	The responder app. `lib/` holds the providers.
`packages/db`	Client factories, generated types, Zod schemas, `roles.ts`.
`packages/api`	Every query and mutation by module. Both apps import from here.
`supabase/migrations`	28 numbered, append-only migrations.
`docs`	Architecture, phases, security, parity, standards, runbook, this manual.

CHAPTER 3

Authorization

> Two independent layers. The **route matrix** decides which screens a role may open. **Row-level security** decides which rows they may touch. The first is clarity; the second is the boundary.

3.1 Silos and helpers

Every user carries a silo and role in JWT app metadata. Three helpers read them — `current_silo()`, `app_role()`, `is_global()` — so a policy reads as the sentence a person would say.

A DISTINCTION WORTH KEEPING

`is_global()` answers "may this role see across departments" and is right for `SELECT`; it returns true for read-only roles such as `Viewer`. Gating writes with it would grant those roles write access, so `can_write_global()` exists separately. Migration 0019 fixed five policies that had conflated the two.

3.2 The roles

The `user_role` enum has 25 values; **23 are assignable to people** from Admin → Users. `System` is an internal service identity, and `Citizen` exists only for the public location-share flow, where a member of the public posts GPS against a single-use token and never holds an account.

GROUP	MEMBERS
`ADMIN_ROLES`	SystemAdmin, SuperAdmin, Admin
`SUPERVISORY_ROLES`	Admin tier + Supervisor, Captain, Chief, FireChief, System
`CALL_TAKER_ROLES`	Admin tier + Operator, ControlRoom, IncidentLogger, System
`DISPATCH_DESK_ROLES`	Admin tier + Dispatcher, Operator, ControlRoom, Supervisor, System
`RESPONDER_ROLES`	Responder, Police, Officer, Metro, Security, Captain, Firefighter, FireChief, Chief, Paramedic, Doctor, Disaster
`POLICE_ROLES`	Captain, Police, Officer, Metro, Security
`FIRE_ROLES`	Chief, FireChief, Firefighter
`MEDICAL_ROLES`	Paramedic, Doctor
`CASE_MANAGER_ROLES`	Admin tier + CaseManager, Dispatcher, Operator, ControlRoom, IncidentLogger, Supervisor, System, Viewer

The two-desk split

Call-taker and dispatch-desk groups barely overlap. An `IncidentLogger` may capture a call and may not work the queue; a `Dispatcher` may work the queue and may not capture calls. This mirrors how the control room is staffed, and it is pinned by tests — a regression merging the two desks would be an authorization change disguised as a convenience.

3.3 The route matrix

`ROUTE_RULES` is an ordered list of path prefixes with permitted roles and optionally a silo. `canAccess()` evaluates it, and it drives the middleware, the sidebar, the top navigation, the command palette and dashboard quick actions — so navigation and enforcement cannot drift.

```
{ prefix: '/dispatch/log-call', roles: CALL_TAKER_ROLES },
{ prefix: '/dispatch',          roles: DISPATCH_DESK_ROLES },
{ prefix: '/police',           roles: [...POLICE_ROLES, ...ADMIN_ROLES], silo: 'MetroPolice' },
{ prefix: '/roster/mine',      roles: undefined },   // every role
{ prefix: '/search' },                               // every role; RLS scopes results
```

3.4 Row-level security

```
create policy incidents_read on public.incidents for select using (
  silo = public.current_silo()                      -- my department owns it
  or public.is_global()                             -- control room / admin
  or public.current_silo() = any(dispatched_silos)  -- routed to us
);
```

That third clause is what makes multi-department response work. Verified against live data: a paramedic searching returns the one call routed to Medical and not the police-only call beside it.

3.5 Privileged functions

Procedures that must bypass RLS run `SECURITY DEFINER` and open with a role check:

```
perform public.require_role('SystemAdmin','SuperAdmin','Admin','Dispatcher',
                            'Operator','ControlRoom','Supervisor','System');
```

> **HISTORIC DEFECT, NOW CLOSED**
>
> Before 0019 every mutating procedure was `DEFINER` with no role check and `EXECUTE` granted to `anon`: a caller with only the public anonymous key could route, dispatch or close any incident in any department, and the audit log would record it as legitimate with a null actor. 0019 added the guards; 0020 fixed revocations that were silently no-ops.

The deliberate exception: `raise_panic()` has no role gate. Any authenticated account may call for help. Gating an emergency on a role matrix is how somebody dies over a provisioning mistake.

VOLUME II

Technology

Every dependency, why it is there, and what actually happens when a page renders, a form submits, an alert fires, or a phone comes back into signal.

CHAPTER 4

The Stack

4.1 Console

PACKAGE	VERSION	WHY
`next`	^15.1.6	App Router. Server components mean incident data is fetched on the server under the user's session and never over-fetched to the client.
`react / react-dom`	^19.0.0	Stable 19. Server actions are the mutation mechanism throughout.
`@supabase/supabase-js`	^2.110.5	Database, auth, storage, realtime client.
`@supabase/ssr`	^0.12.3	Cookie-based session handling for server components. **Must stay version-paired** with supabase-js.
`zod`	^3.23.8	Input schemas shared with the mobile app via `packages/db`.
`react-hook-form`	^7.53.1	The few genuinely complex client forms; most use uncontrolled server-action forms.
`@hookform/resolvers`	^3.9.0	Bridges Zod into react-hook-form.
`@tanstack/react-query`	^5.59.0	Client-side caching for the live boards.
`@tanstack/react-table`	^8.20.5	Sorting and column logic for the larger registers.
`leaflet`	^1.9.4	Maps. Chosen over a hosted SDK because tiles come from Esri/OSM and there is no per-view billing.
`recharts`	^2.13.0	Dashboard, KPI and analytics charts.
`pdf-lib`	^1.17.1	Case-report and fire-report PDFs, including embedding scene photographs.
`lucide-react`	^0.451.0	Icon set.
`@radix-ui/*`	various	Dialog, dropdown, label, select, slot, toast — accessible primitives rather than bespoke ones.
`tailwindcss`	^3.4.13	Styling, with CSS variables driving the eight themes.
`clsx · tailwind-merge · cva`	—	Conditional and variant class composition.
`undici`	^8.9.0	HTTP client for the server-side proxies.
`server-only`	^0.0.1	Build-time guard: importing a server module from a client bundle fails the build rather than leaking a key.

4.2 Responder app

PACKAGE	VERSION	WHY
`expo`	~52.0.0	SDK 52. New Architecture disabled deliberately — it destabilised startup on this dependency set.
`react-native`	0.76.9	Paired with SDK 52.
`react`	18.3.1	RN 0.76 is not on React 19.
`expo-router`	~4.0.11	File-based routing; tabs plus a stack.
`expo-notifications`	~0.29.14	OS push. Every call guarded; Android needs FCM (see 41.3).
`expo-location`	~18.0.4	Duty heartbeat, panic position, nearby sorting.
`expo-image-picker`	~16.0.3	Camera and library evidence capture.
`expo-av`	~15.0.2	Voice-note recording.
`expo-file-system`	~18.0.12	Retaining attachments out of the OS cache so evidence survives eviction.
`expo-haptics`	~14.0.1	Escalating feedback while the panic button arms.
`@react-native-community/netinfo`	11.4.1	Connectivity, so the outbox does not burn its retry budget offline.
`@react-native-async-storage/async-storage`	1.23.1	Session persistence and the durable outbox.
`expo-updates`	~0.27.5	Over-the-air JavaScript updates.
`expo-local-authentication`	~15.0.2	Available for device-lock work; not yet surfaced in the UI.
`react-native-url-polyfill`	^2.0.0	supabase-js needs WHATWG URL, which RN lacks.

4.3 Shared packages and tooling

ITEM	PURPOSE
`@criticom/db`	Client factories (browser, server, bearer, service-role), hand-authored database types, Zod schemas, the route matrix.
`@criticom/api`	Every query and mutation, grouped by module. The only place either app talks to the database.
`turbo`	^2.1.3 — task orchestration and caching across the workspace.
`pnpm`	9.12.0 — workspace package manager. Root overrides pin a single `@types/react`.
`biome`	1.9.4 — linting. Formatter not enforced; see 44.3.
`typescript`	^5.6.3 — strict, with `noUncheckedIndexedAccess` on mobile.
`node`	>=22 — the test runner uses `--experimental-strip-types`.

A HAND-AUTHORED TYPES FILE

`packages/db/src/types.generated.ts` is written by hand despite its name. supabase-js 2.110 requires `Views`, `CompositeTypes` and `Relationships` keys or every table resolves to `never`. Regenerate with `supabase gen types typescript` only if you are prepared to re-add those; otherwise edit it alongside the migration.

CHAPTER 5

How a Request Works

5.1 Rendering a page

1. Request hits **middleware**, which reads the session cookie and evaluates the route matrix. Unauthorized → `/403`. Unauthenticated → `/login`. Static assets are excluded, or the logo 404s.
2. The `(app)` **layout** loads: session, operational settings, ops links, department branding. All four fail safe.
3. The **page** (a server component) calls functions in `@criticom/api` with a server Supabase client carrying the user's cookies.
4. Postgres applies **RLS** and returns only permitted rows.
5. HTML is streamed. Client components — topbar, alert centre, assistant dock, maps — hydrate afterwards.

Every page is `force-dynamic`. Caching a page rendered for one operator's permitted view and serving it to another is exactly the failure this system cannot have.

5.2 Submitting a change

1. A `<form action={someAction}>` posts to a **server action**.
2. The action calls `getSession()` and re-checks the role. **Never trust that the page was reachable** — a form can be posted directly.
3. It calls a function in `@criticom/api`, which either writes tables directly (RLS applies) or calls a stored procedure (guard applies).
4. `logAudit` records it.
5. `revalidatePath` refreshes the affected screens.

5.3 Realtime

Boards, timelines and the alert bell subscribe to Postgres change streams. RLS applies to the socket too, so a subscriber is only delivered rows they could have queried. Each subscription gets its own channel topic, because Supabase reuses a channel per topic and rejects listeners added after `subscribe()`.

5.4 The offline path

1. The responder acts. The operation is written to **AsyncStorage** with the time it was performed.
2. The UI updates from the queue, not from the server — so the screen shows what was asked for.
3. Delivery is attempted only when NetInfo reports the internet reachable.
4. On success the entry is removed; on failure it backs off, and after eight connected failures it parks as *failed* and waits for a human.

5. Handlers carry `occurredAt`, so the server records when the crew acted rather than when the network returned.

VOLUME III

Operational Flow

The path a call takes from the telephone to a closed case file, the statuses it passes through, and how people are told about it.

CHAPTER 6

The Life of a Call

#	STEP	SCREEN	WHAT HAPPENS
1	**Call arrives**	`/dispatch/log-call`	Call-taker captures caller, description, classification and location. An intake clock runs from the moment the screen opens.
2	**Duplicate check**	same	`check_duplicate_incident()` looks back 12 hours by caller number, and by location + type.
3	**Incident created**	same	`log_incoming_call()` writes the row, allocates `INC-YYYYMMDD-NNNN`, audits. Call-taker, category and region are stamped.
4	**Routed**	same, or `/dispatch/pending`	`route_incident()` sets receiving departments, appends timeline, audits and raises one alert each — one transaction.
5	**Crews notified**	everywhere	Realtime to open screens; High/Critical raise a blocking modal; Expo push to locked phones.
6	**Unit assigned**	`/dispatch/assign/[id]`	`dispatch_unit()` commits an appliance and removes it from the available pool.
7	**Crew responds**	phone, or `/responder`	En Route → Arrived → On Scene → Transporting → Resolved. Each queues locally and carries the time pressed.
8	**Scene recorded**	phone incident detail	Photographs and voice notes retained on device, uploaded to `incident-media`, filed against the incident.
9	**Report filed**	department section	Police case, fire report, or patient care record.
10	**Closed**	`/incidents/[id]`	`close_case()` records the reason and audits. The PDF embeds timeline, attachments and audit trail.
11	**Watched throughout**	—	If any step stalls, the automation engine escalates without anybody asking.

CHAPTER 7

The Status Model

STATUS	SET BY	MEANING
New / Pending	System	Created, not yet in a queue state. Rare in normal operation.
Pending Dispatch	`log_incoming_call()`	Waiting for a department. **This is what the SLA clock counts.**
Pending Analysis	`log_incoming_call()`	The Disaster Management equivalent — assessment before deployment.
Dispatched	Dispatch desk	Routed to departments, and possibly a specific unit.
En Route	Responder	Crew is moving. Sets `acceptance`.
Arrived / On Scene	Responder	Stamps `arrived_timestamp` once, and never moves it again.
Transporting	Responder	Patient or subject in transit.
Active	Supervisory	A long-running incident being managed.
Resolved	Responder	Work complete on scene; the case file may still be open.
Closed	`close_case()`	Reviewed and closed with a reason. Terminal.

> **WHY ARRIVAL IS WRITE-ONCE**
>
> `arrived_timestamp` is set only when it is null. Response-time KPIs and any inquiry into a bad outcome are computed from exactly that column, so a later status change — or a replayed offline update — must not be able to rewrite it.

Who may advance a status

`update_responder_status()` admits the responder roles plus supervisory and dispatch tiers. The forward ladder offered in the interface is `Dispatched → En Route → Arrived → On Scene → Transporting → Resolved`; one step at a time, because skipping states loses the timings the ladder exists to capture.

CHAPTER 8

The Alert Model

8.1 Anatomy

FIELD	MEANING
`silo`	Which department's copy this is. A fanout writes one row per department.
`target_user_id`	Set for a personal alert; null for a department broadcast.
`severity`	Low · Medium · High · Critical.
`presentation`	`toast` — corner notice. `modal` — blocking interrupt.
`requires_ack`	Acknowledgement recorded in `alert_reads`.
`expires_at`	Auto-hide. An aged-out alert is not something still to act on.
`created_by_name`	A person, or `Criticom automation` — so an automatic escalation is always distinguishable from a colleague's.

8.2 Delivery

CHANNEL	COVERS	BLIND SPOT
Realtime	Everything, instantly, while a screen is open.	Closed or backgrounded app.
Expo push	Locked phones.	Needs the native module, and FCM on Android.
Local notification	Alert arriving over realtime while the app is backgrounded.	App fully closed.
Inbox	Everything after the moment — what came in while you were busy.	Nothing; it is the durable record.

The overlap is deliberate. Either channel alone has a blind spot and a missed dispatch is not a recoverable error; duplicates are cheap by comparison and are deduplicated by alert id.

8.3 Tuning

Admin → Operations Settings controls which severity interrupts with a blocking pop-up (`alerts.modal_min_severity`), whether a sound plays, and how long a toast stays. Changing these changes what crews are interrupted for — treat it as an operational decision, not a preference.

VOLUME IV

Role Playbooks

One playbook per desk. What you can reach, what your shift looks like, every task you perform, and what you cannot do — with who to ask instead.

CHAPTER 9

How to Read a Playbook

> Twelve playbooks cover all 23 assignable roles. Several roles share a playbook because they share a desk — a `Police`, an `Officer` and a `Metro` see the same screens and do the same work; the distinction is rank, not permission.

9.1 Structure

SECTION	WHAT IT TELLS YOU
Who this is	The job, and which role values map to it.
Reach	Exactly which screens the route matrix opens for this role. Anything not listed redirects to `/403`.
Shift routine	The order to do things in, start to end.
Tasks	Numbered procedures. Full steps are in Volume V; the playbook says which apply to you and in what order.
Not yours	What this role deliberately cannot do, and who to go to.

9.2 Two rules that apply to everybody

SILO SCOPE

Unless you hold a global role, you see your own department's records plus anything routed to your department. This is enforced in the database, so it holds on every screen, every export and every search — there is no view that shows you more.

EVERYTHING YOU DO IS ATTRIBUTED

Every mutation writes an audit row with your name, department and a description. This is not surveillance; it is what makes the record usable as evidence, and it is what protects you when somebody asks why a decision was made.

CHAPTER 10

Call-Taker

> **IncidentLogger**
> The intake desk. Answers the telephone and turns a call into an incident record. Does not dispatch.

10.1 Reach

SCREEN	USE
`/dashboard`	Your board.
`/dispatch/log-call`	**Your primary screen.**
`/incidents` · `/incidents/[id]` · `/incidents/map`	Look up what you logged.
`/reports` · `/report-hub` · `/reports/builder`	Reporting.
`/case-manager`	Case review.
`/cameras` · `/cameras/footage`	Camera wall and retrieval.
`/hierarchy`	Reporting lines.
`/roster/mine` · `/roster/allocation`	Your schedule and the duty board.
`/leave`	Apply for leave.
`/performance/mine`	Your own throughput.
`/notifications` · `/profile` · `/search` · `/assistant`	Universal.

10.2 Shift routine

1. Sign in. Check the top bar shows your department and the alert bell for anything outstanding.
2. Open **Dispatch → Log Incoming Call** and leave it open. The intake clock starts when the screen opens, so a fresh screen per call keeps the timing honest.
3. Take calls (10.3).
4. Between calls, check **Incidents** for anything you logged that has not moved.

5. At end of shift, confirm nothing you captured is still sitting in the queue unrouted, and hand over anything unusual verbally.

10.3 Tasks

Log an incoming call

Screen /dispatch/log-call · **Writes** incidents, audit_logs · **Full procedure** 22.1

1. Capture **Caller name** and **Contact** as the caller gives them. Do not correct spelling later at the cost of listening.
2. Enter the **Call description / narrative** — what is happening, in their words.
3. Set the **location**: type the address and take a suggestion, or drop a pin on the map. The two stay in step; you do not need to type coordinates.
4. Classify: **incident type** from the configured category tree, then **Region** and **Incident severity**.
5. If casualties: **Number of victims** and, for EMS, the **Medical severity code** (P1 Red through P4 Blue). Tick *EMS not required* if it plainly is not a medical call.
6. Watch for the **duplicate warning**. If it names an existing incident, stop and check before creating a second one.
7. Choose the **receiving departments** if you are permitted to route; otherwise submit and the dispatch desk will.
8. Submit. You get the incident number in a confirmation.

AFTERWARDS The incident exists with a sequential number, your name stamped as call-taker, and — if routed — an alert on every screen in the receiving departments and a push to their phones.

Attach something to a call at intake

Screen /dispatch/log-call · **Writes** incident_attachments, storage

1. Use the attachment control before submitting.
2. Submit as normal.

IF THE UPLOAD FAILS The incident is still created and you are told the attachment did not upload. Re-attach it from the incident page — the call is never lost to a failed file.

Ask a caller for their GPS position

Screen /incidents/[id] · **Needs** BulkSMS configured · **Full procedure** 22.5

1. Open the incident.
2. Enter the caller's mobile number in **Contact citizen** and submit.
3. They receive an SMS with a single-use link. When they open it and allow location, the position lands on the incident.

USE WHEN The caller cannot describe where they are — common on freeways and in informal settlements.

10.4 Not yours

YOU CANNOT	GO TO
Work the pending queue or assign units	Dispatcher (Chapter 12). This is the two-desk split and it is deliberate.
Change an incident's status	The attending crew, or the dispatch desk.
Close a case	Case manager or supervisor.
Open a department's reports	That department.

CHAPTER 11

Control Room Operator

> **ControlRoom · Operator**
> The full control-room desk: takes calls *and* dispatches. Global silo scope — sees every department.

11.1 Reach

Everything the call-taker reaches, **plus** the whole dispatch desk: `/dispatch/pending`, `/dispatch/assign/[id]`, `/dispatch/kpi`, `/dispatch/ptt`, plus `/disaster`. Not the departmental report sections, and not `/admin`.

11.2 Shift routine

1. Sign in and clear the alert bell — read anything raised overnight, including the morning digest.
2. Open **Dispatch → Pending Queue**. The figure at the top is your queue depth; it is live.
3. Work the queue oldest-and-worst first. Severity, then call time, is how it is already sorted.
4. Take calls as they come (10.3), routing them yourself rather than leaving them queued.
5. Watch for automatic escalations — an unassigned call raises a High alert at 10 minutes and a blocking Critical at 20.
6. Check **Dispatch → KPI** before handover so you hand over a described state, not a surprise.

11.3 Tasks

Route a call to departments

Screen /dispatch/pending or the incident · **Procedure** `route_incident()` · **Full** 22.2

1. Open the incident from the queue.
2. Select every department that needs to attend. A structure fire is usually Fire *and* EMS; add Police if a cordon is needed.
3. Add dispatch notes if there is something the crew must know before arriving.
4. Confirm.

AFTERWARDS Status becomes Dispatched, `dispatch_time` is stamped, the timeline records who routed it, and every screen and phone in those departments is alerted. High and Critical open a blocking pop-up crews must acknowledge.

Assign a specific unit

Screen /dispatch/assign/[id] · **Procedure** `dispatch_unit()` · **Full** 22.3

1. Open the incident from the queue and choose **Assign**.
2. Review the map, the incident detail and the available units with their road-routed ETAs.
3. Select the unit and confirm.

AFTERWARDS The appliance leaves the available pool and the incident carries the unit. Routing tells a department; assigning commits an appliance — do both.

Send a message to a radio

Screen /dispatch/ptt · **Needs** RealPTT configured · **Full** 28.3

1. Open the PTT console.
2. Enter the destination account and the message text.
3. Send. The two-hour audio log is on the same screen.

11.4 Not yours

Administration (`/admin`), departmental report capture, and HR approvals. For a role or fleet change, go to an administrator; for leave decisions, a supervisor.

CHAPTER 12

Dispatcher

> **Dispatcher**
> The dispatch desk only. Works the queue, allocates units, watches the board. Does not take calls.

12.1 Reach

The dispatch desk (`/dispatch/pending` , `/dispatch/assign/[id]` , `/dispatch/kpi` , `/dispatch/ptt`), incidents, reports, case management, cameras, hierarchy, roster allocation, leave, own performance. **Not** `/dispatch/log-call` — intake is the call-taker's desk.

> **WHY YOU CANNOT LOG A CALL**
> The two-desk split matches how the control room is staffed and keeps the operational KPIs meaningful: call-taker throughput and dispatcher throughput are separate measures, and they stop being measures if one person does both under one role.

12.2 Shift routine

1. Sign in; clear the alert bell.
2. **Dispatch → Pending Queue** is your screen for the shift. Leave it open — it is live.
3. For each call: route to departments, then assign a unit.
4. Watch the dashboard board for crews that have gone quiet — the engine will flag no-arrival after 30 minutes and silence after 4 hours, but you will usually see it first.
5. Use the PTT console rather than a personal handset, so the traffic is logged.
6. At handover, walk the incoming dispatcher through anything still open.

12.3 Tasks

Routing (11.3), assigning (11.3) and PTT (11.3) as above, plus:

Respond to an SLA escalation

Trigger automation rule `sla_unassigned` · **See** 37.2

1. A High alert names the incident and how long it has waited. At the critical tier it opens a blocking pop-up.
2. Open the incident from the alert.
3. Either assign it, or record in the timeline *why* it cannot be assigned — no available unit, awaiting mutual aid.

NOTE The alert fires once per tier, not repeatedly. If you see the same incident escalate twice it has crossed from warn to critical, which is a real deterioration.

Handle an officer-emergency

Trigger panic from the responder app · **See** Chapter 32

1. A Critical blocking alert opens naming the officer. Acknowledge it.
2. A red banner stays at the top of every page until the incident is closed — that is your standing reminder.
3. Open the incident. The address is the officer's coordinates, or says plainly that the position is unknown.
4. Assign the nearest units. Raise it on the radio in parallel; do not rely on the platform alone.
5. Close the incident only once contact is made, with a note recording the outcome.

IF IT WAS ACCIDENTAL Close it as any other incident, with a note. The button is deliberately easy to trigger — an acknowledged false alarm is a far better failure than hesitation.

CHAPTER 13

Field Responder

> **Responder**
> The generic field role. Attends incidents, records what happened. Most crews will hold a departmental role instead (Chapters 14–17), which adds that department's reports.

13.1 Reach

`/responder`, `/shift/*`, `/incidents`, `/reports`, `/roster/mine`, `/roster/allocation`, `/leave`, `/training`, `/performance/mine`, plus the responder app. `Responder` is additionally admitted to all three departmental sections, which is unusual — most field roles get only their own.

13.2 Shift routine

1. **Complete start of shift before anything else** (13.3). You cannot take incidents until it is done, on the phone or the console.
2. Go on duty on the phone's Profile tab. This starts the GPS heartbeat so dispatch can see you.
3. Work assigned incidents from the Incidents tab.
4. Record evidence at the scene while you are there.
5. Before signing out, check the sync tray is empty.
6. Sign out from the phone — this releases your vehicle and clears your duty status.

13.3 Tasks

Start of shift

Screen Shift tab, or /shift/mental-check · **Writes** mental_checks, apparatus, vehicle_checklists · **Full** 24.1

1. **Mental readiness.** Answer all six honestly: memory recall, judgment, situational awareness, not under the influence, anger check, reflexes. Add comments if anything is borderline.
2. **Vehicle selection.** Choose your appliance from those available in your department.
3. **Vehicle inspection.** Record fuel level (¼, ½, ¾, Full) and confirm lights, tyres, equipment stock, communications and brakes.

WHY IT GATES EVERYTHING The mental check is a fitness-for-duty declaration and the inspection is a roadworthiness record. Attending a call with neither on file means there is no record you were fit to attend — which is the record that matters if the call goes wrong.

Advance an incident status

Screen phone Incidents tab, or /responder · **Procedure** `update_responder_status()` · **Full** 23.1

1. Open the incident.
2. Press the single forward button — it always shows the next status: **Mark En Route**, then Arrived, On Scene, Transporting, Resolved.
3. Press it **when the thing actually happens**, not when you next have a moment.

WITH NO SIGNAL The card changes immediately and shows *Waiting to send*. The time recorded is when you pressed it, not when signal returned. Do not press it again.

Record scene evidence

Screen phone incident detail · **Writes** incident_attachments, storage · **Full** 30.4

1. Open the incident, scroll to **Scene evidence**.
2. **Camera** to take a photograph, **Library** to attach an existing one, or **Record voice note** for a spoken account.
3. Capture what you need. Coverage does not matter — files are stored on the phone and upload themselves.

AFTERWARDS Attachments appear in the case file and in the case-report PDF. A photograph of a scene cannot be retaken, so capture generously.

Add a timeline update

Screen phone incident detail, or /incidents/[id]

1. Type into the update box and send.

USE FOR Anything the record should carry that a status cannot express: what you found, what you did, who else attended, why something took the time it did.

13.4 Not yours

Routing and assigning, closing cases, approving leave, administration. If you need a call sent to another department, tell dispatch — do not try to route it yourself.

CHAPTER 14

Metro Police Officer

> **Police · Officer · Metro · Security · Captain**
>
> Metro Police field roles. Everything in Chapter 13, plus the police report suite. Captain additionally holds supervisory reach.

14.1 Reach

Field responder reach, **plus** `/police` and everything under it — reports, firearms, security (SIRA) reports, crime hotspots. Restricted to the `MetroPolice` silo: an officer in another department cannot open it even with the same role value.

14.2 Tasks

File a police case report

Screen /police/reports/new · **Writes** metro_reports · **Full** 25.1

1. Open **Police → Reports → New**.
2. Head the report: **case number**, **date** and **time of offence**, **location**, **station**, **incident category**.
3. Record yourself: **officer name**, **badge number**, **partner name**, and whether **bodycam** was active.
4. Record the people: **suspect** name and description, **offender** name and ID, **victim** name, ID, contact and injuries.
5. Record the outcome: **arrest made**, **charges**, **use of force**, **fine amount** or **ticket number**.
6. Record property: **vehicle** details and impound, **weapon** type and size, **ammunition rounds**, **contraband**, and the **evidence log** (weapons, drugs, property).
7. Write the **officer narrative** — the account that will be read in court.
8. Submit.

AFTERWARDS The report is filed against your department and appears in the archive and in Report Hub exports. Link it to an incident where one exists.

Issue a firearm authorization

Screen /police/firearms/new · **Writes** firearm_authorizations

1. Open **Police → Firearms → New**.
2. Record the officer, the firearm and the authorization period.
3. Submit.

ON RETURN Use **RETURN** against the register entry. An unreturned authorization is a live accountability gap.

Capture a SIRA security-guard report

Screen /police/security/new · **Writes** security_reports

1. Open **Police → Security → New**.
2. Record the guard: name, guard ID, badge, contact, ID number, **SIRA number** and **grade**.
3. Record whether a **firearm was issued** and its number.
4. Record GPS arrival and departure times and the site.
5. Submit.

TO CLOSE IT Use **RESOLVE** and enter a resolution summary.

Check crime hotspots

Screen /police/crime-hotspots · **Reads** crime_stats

1. Open **Police → Crime Hotspots**.
2. Filter by station and category. The dataset is the SAPS statistics, ranked.

USE FOR Deployment planning and briefings, not for live response.

CHAPTER 15

Firefighter

> **Firefighter · Chief · FireChief**
>
> Fire & Rescue field roles. Chapter 13, plus the fire report suite, the hydrant register and post-incident review. Chief and FireChief additionally hold supervisory reach.

15.1 Reach

Field responder reach, **plus** `/fire` — reports, hydrants, PIR. Restricted to the `MetroFire` silo.

15.2 Tasks

File a fire incident report

Screen /fire/reports/new · **Writes** firefighter_reports · **Full** 25.2

1. Open **Fire & Rescue → Reports → New**.
2. Head it: **report number**, **date** and **time of incident**, **location**, **type of incident**, **station number**, **fire department name**, and the **reporting officer** name, rank and ID.
3. Record the **timings** in minutes: dispatch, arrival, clear, total. These feed the response-time measures.
4. Record the property: **type**, **owner**, **contact**, **value**, **contents value**, **insurance**, **damage estimate**.
5. Record the fire: **point of origin** (room and object), **ignition source**, **cause**, **initial observations**, **smoke conditions**, **flame spread**, **heat signature**.
6. Record the response: **suppression details**, **equipment used**, **ventilation**, **salvage and overhaul**, **utilities secured**, **search and rescue**, **persons rescued** with location and condition.
7. Record casualties separately for **civilians** and **firefighters** — injuries and fatalities, each with detail.
8. Record **hazmat** operations, **hazards identified**, **medical aid provided**, **treatment**, **mutual aid** and **other agencies on scene**.
9. Close with the **narrative summary**, **challenges faced**, **successes**, **recommendations**, and whether a **fire investigation** or **post-incident critique** is to follow.
10. Submit.

AFTERWARDS Download the PDF from the report page for the station file. If a critique is scheduled, the PIR flow follows.

Publish a post-incident review

Screen /fire/pir · /fire/pir/publish/[reportId] · **Writes** public_incident_reports

1. Open **Fire & Rescue → PIR** and choose the report.
2. Review the content that will be published.
3. Publish.

CARE This makes a summary publicly visible. Check names and addresses before publishing.

Use the hydrant register

Screen /fire/hydrants · **Reads** hydrants

1. Open **Fire & Rescue → Hydrants**.
2. Find the incident location on the map; hydrants render as markers with their detail.

USE FOR Pre-planning and live water supply. This is operational data during a fire, not a maintenance list.

CHAPTER 16

Paramedic

> **Paramedic · Doctor**
> EMS field roles. Chapter 13, plus the patient care record and ambulance restocking.

16.1 Reach

Field responder reach, **plus** `/medical` — PCR create, view and edit, and restock. Restricted to the `Medical` silo.

16.2 Tasks

Complete a patient care record

Screen /medical/pcr/new · **Writes** paramedic_reports and its child tables · **Full** 25.3

1. Open **EMS → PCR → New**.
2. **Patient:** name, date of birth, gender, address, contact, insurance, next of kin, allergies.
3. **History:** presenting complaint and relevant medical history.
4. **Assessment:** your findings.
5. **Vital signs:** record each set with its time. Multiple sets are expected on a transport.
6. **Interventions:** what you did, and when.
7. **Handover:** the receiving facility and the legal handover record.
8. **Signatures:** capture responder, hospital representative and doctor on the signature pad.
9. Submit.

EDITING Use **PCR → EDIT** to correct or complete a record. It remains linked to the incident, and every change is audited.

Record a restock

Screen /medical/restock · **Writes** ambulance_restocks

1. Open **EMS → Restock**.
2. Select the **unit**, enter the **drug name**, **quantity used** and **quantity remaining**.
3. Submit.

DO IT AT THE END OF EACH CALL, not at end of shift. Quantity remaining is what the next crew relies on.

CHAPTER 17

Disaster Management Officer

> **Disaster**
> Major incidents, floods, multi-agency response. Works over longer timescales than the other desks.

17.1 Reach

Field responder reach, **plus** `/disaster` — monitor, dispatch, partners, post-mortem. Restricted to the `EthekwiniDisaster` silo. Calls arrive as **Pending Analysis** rather than Pending Dispatch, because the first step is assessment.

17.2 Tasks

Dispatch resources to a disaster

Screen /disaster/dispatch · **Writes** disaster_logs

1. Open **Disaster → Dispatch**.
2. Select the resources and the partners required.
3. Confirm.

Post a periodic update

Screen /disaster · **Writes** disaster_logs

1. Find the log entry for the incident.
2. Add the update text and submit.

CADENCE MATTERS A disaster log with hourly entries is a usable account afterwards; one with two entries is not. The engine will nudge after 4 hours of silence, but do not wait for it.

Write a post-mortem

Screen /disaster/post-mortem · **Writes** post_mortem_reports

1. Open **Disaster → Post-mortem**.
2. Record what happened, what worked, what did not, and the recommendations.
3. Submit.

Find a mutual-aid partner

Screen /disaster/partners · **Reads** mutual_aid_partners

1. Open **Disaster → Partners**.
2. Filter by the resource you need. 41 partners are registered with their contacts and capabilities.

CHAPTER 18

Supervisor

> **Supervisor · Captain · Chief · FireChief**
>
> The supervisory tier. Runs people: rosters, duty allocation, leave, training, performance — and holds the dispatch desk too.

18.1 Reach

The widest non-administrative reach: `/hr`, `/roster` and all of it, `/leave` with decisions, `/training`, `/performance` in full, `/analytics`, `/hierarchy`, the dispatch desk, incidents, reports, cameras. Captain, Chief and FireChief also keep their departmental section from Chapters 14–16.

18.2 Routine

1. **HR Command Centre** (`/hr`) first: headcount, on duty, on leave today, approvals waiting, certifications expiring within 30 days.
2. Decide any pending leave — each request shows its coverage impact.
3. Check the duty board for gaps before the shift starts.
4. Chase unacknowledged allocations.
5. Weekly: performance reports and the rollup.
6. Annually from 1 July: appraisal plans.

18.3 Tasks

Decide a leave request

Screen /hr or /leave · **Writes** leave_entries, alerts, audit_logs · **Full** 26.3

1. Open **HR** — pending requests are listed with an inline decision.
2. Read the **coverage impact**: the percentage of the department away if you grant it.
3. Approve or reject.

AFTERWARDS The member gets a targeted alert immediately, and the decision is recorded against your name with a timestamp.

Allocate a duty

Screen /roster/allocation · **Writes** rosters · **Full** 26.2

1. Open **Roster → Allocation**. The board is post-first: every post the shift must fill, with gaps visible.
2. Assign a member to a post, setting **vehicle**, **sector**, **call sign** and any **special instructions**.
3. Repeat until the gaps are closed.
4. **Publish the day.** Until you publish, allocations are drafts and members cannot see them.
5. Use **Copy forward** to seed the next day from this one.

THEN CHASE ACKNOWLEDGEMENTS. Publication makes it visible; acknowledgement is the accountability half. The engine reminds members 12 hours before the start.

Record training and certifications

Screen /training · **Writes** training_records, certifications

1. Open **Training**.
2. For training: select the member, enter the **training name**, **date completed** and **notes learned**.
3. For a certification: select the member, enter the **name**, **expiry date**, **issued by**, **date achieved** and remarks.

THE EXPIRY DATE IS LOAD-BEARING. It drives automatic warnings at 30, 7 and 1 days to both the member and the department. A certification with no expiry recorded warns nobody.

Run the annual appraisal cycle

Screen /performance/* · **Writes** performance_plans, performance_scores · **Full** 26.5

1. **Define KPIs** for the cycle year: title, description, weight, department.
2. **Generate plans** for the cycle year — one per member.
3. At mid-year, open each plan and **save a score** per KPI with comments and evidence.
4. **Approve the stage** to lock it.
5. Repeat at final.

CYCLE The year runs from 1 July.

Set who reports to whom

Screen /hierarchy/person · **Writes** reporting_lines

1. Open **Hierarchy** and find the member.
2. Set their reporting line.

WHY IT MATTERS The performance rollup aggregates along these lines. A missing line means somebody's numbers do not reach their manager.

CHAPTER 19

Case Manager

> **CaseManager**
> Reviews and closes resolved incidents, and produces the official case file.

19.1 Reach

`/case-manager` and its audit view, `/incidents`, `/reports`, `/report-hub`, `/hierarchy`, `/performance` rollup and reports, `/roster/allocation`, `/leave`.

19.2 Tasks

Close a case

Screen /incidents/[id] or /case-manager · **Procedure** `close_case()` · **Full** 23.4

1. Open the incident and read the timeline end to end.
2. Check the departmental report exists where one is required.
3. Check the attachments are present.
4. Enter a **reason** and close.

AFTERWARDS Status becomes Closed, `closed_at` is stamped, the reason goes into the timeline, and an audit row records you closed it. Closing is not reversible through the interface.

Produce the case-report PDF

Screen /incidents/[id] · **Endpoint** /api/pdf/incident/[id]

1. Open the incident and choose the PDF download.
2. Wait — the document downloads and embeds every scene photograph, so it is the slowest request the platform serves.

CONTENTS Incident detail, full timeline, embedded attachments and the audit trail. It is complete as a single document, which is what a case file has to be.

Review the audit trail

Screen /case-manager/audit · **Reads** audit_logs

1. Open **Case Manager → Audit**.
2. Filter to the period or entity in question. Every mutation carries actor, department, action and detail.

CHAPTER 20

Staff and Viewer

Staff

Non-operational employees. Self-service only.

Reach: `/dashboard`, `/profile`, `/profile/my-hr`, `/roster/mine`, `/roster/allocation`, `/leave`, `/reports/builder`, `/notifications`, `/search`.

Tasks: apply for leave (26.3), view your schedule, keep your profile current, check your HR record.

Viewer

Read-only oversight. Sees across departments and changes nothing.

Reach: incidents, reports, report hub, case management, cameras, hierarchy, analytics, disaster, performance rollup and reports. **Every screen is read-only** — the role appears in `is_global()` but deliberately not in `can_write_global()`.

WHO THIS IS FOR

Oversight that needs the whole picture without the ability to alter it: senior command, auditors, a municipal oversight function. If somebody needs to change something, they need a different role, not an exception.

CHAPTER 21

Administrator

> **Admin · SystemAdmin · SuperAdmin**
>
> Runs the platform itself. Reaches every screen, including all of `/admin`. SystemAdmin and SuperAdmin additionally keep the left sidebar, because they genuinely work across every module at once.

21.1 Routine

1. **Admin → System Status** daily. Each check says what it means rather than reporting a number.
2. **Admin → Automations** — confirm the engine ticked and nothing is erroring.
3. Weekly: integrations probe, fleet and device registers, audit spot-check.
4. On joiners and leavers: create or retire accounts the same day.

21.2 Tasks

Full procedures are in Volume VII. In brief:

TASK	SCREEN	SEE
Set a user's role	`/admin`	34.1
Add personnel	`/admin/personnel`	34.2
Add a control-room operator	`/admin/operators`	34.3
Add or retire an appliance	`/admin/vehicles`	34.4
Register a device	`/admin/devices`	34.5
Add an incident category or leave type	`/admin/config`	35.1
Repoint an external ops link	`/admin/config`	35.2
Tune an SLA clock or alert behaviour	`/admin/operations`	36
Tune or disable an automation	`/admin/automations`	37.3
Enable an integration	`/admin/integrations`	38.2
Set department branding	`/admin/departments`	39
Check platform health	`/admin/status`	43.1

TWO THINGS AN ADMINISTRATOR MUST UNDERSTAND

- **Role changes are silo-scoped in effect.** Giving somebody `Paramedic` in the MetroPolice silo gives them a role that opens no departmental section, because `/medical` requires the Medical silo. Set both.
- **Disabling an automation is a real operational decision.** Turning off unassigned-call escalation means nobody is told when a call ages. Do it knowingly, and turn it back on.

VOLUME V

Task Procedures

Every task on the console, in full. Each one names the screen, who may do it, what it writes to the database, and what happens afterwards.

CHAPTER 22

Intake and Dispatch

22.1 Log an incoming call

Log an incoming call

Screen /dispatch/log-call · **Who** Admin tier, Operator, ControlRoom, IncidentLogger · **Procedure** `log_incoming_call()`

1. **Open the screen fresh for each call.** An intake clock starts on open and is shown as *Time since call intake started*. It turns amber then red at the thresholds set in Operations Settings.
2. **Caller name** and **contact number**.
3. **Call description / narrative** — what is happening.
4. **Location.** Either type into the address field and pick a suggestion, or click the map. Address, **Latitude** and **Longitude** stay in step automatically. Add **Suburb**, **City** and **Postal code** where known.
5. **Classification.** Pick the incident category and sub-category from the configured tree, then **Region** (Central, North, South, West, Outer).
6. **Incident severity** — Low, Medium, High or Critical. This decides whether receiving crews get a corner toast or a blocking interrupt.
7. **Casualties**, if any: **Number of victims** (1–5, 6-10, 10+) and **Medical severity code** (P1 Red Immediate, P2 Yellow Urgent, P3 Green Delayed, P4 Blue Expectant). Tick *EMS not required* when it is plainly not a medical call.
8. **Internal dispatch notes** — anything the crew must know before arriving.
9. **Check the duplicate warning.** If it names an existing incident, open it rather than creating a second.
10. **Receiving departments**, if you may route. Selecting none files the incident under your own department for the dispatch desk to route.
11. Submit.

AFTERWARDS An incident row exists with a sequential `INC-YYYYMMDD-NNNN` number, status *Pending Dispatch* (or *Pending Analysis* for Disaster Management), your name as call-taker, and an audit row. If you routed it: status *Dispatched*, alerts on every screen in those departments, and a push to their phones.

22.2 Route an incident to departments

Route an incident

Who Admin tier, Dispatcher, Operator, ControlRoom, Supervisor · **Procedure** `route_incident()`

1. Open the incident from **Dispatch → Pending Queue**.
2. Select every department that must attend — more than one where the call needs it.
3. Add notes if required.
4. Confirm.

ONE TRANSACTION Status → Dispatched; `dispatch_time` stamped; `dispatched_silos` set (which is what grants those departments visibility); your name recorded as dispatcher; a timeline entry; an audit row; and one alert per department. It cannot half-succeed.

22.3 Assign a unit

Assign a specific appliance

Screen /dispatch/assign/[id] · **Procedure** `dispatch_unit()`

1. From the queue, choose **Assign** on the incident.
2. Review the incident detail, the map, and the available units with road-routed ETAs.
3. Select a unit and confirm.

AFTERWARDS The appliance leaves the available pool; the incident carries `unit_id` and `unit_type`. Routing tells a department; assigning commits an appliance.

22.4 Handle a duplicate

Check for a duplicate

Automatic on the log-call screen · **Procedure** `check_duplicate_incident()`

1. The check runs as you capture, over a 12-hour window, matching on **caller number**, and separately on **location + incident type**.
2. If a match is shown, open it and confirm whether it is the same event.
3. Same event — add what the new caller told you as a timeline update on the existing incident and do not submit a new one.
4. Different event — continue and submit.

WHY IT MATTERS Two incidents for one crash sends two crews and produces two partial records of the same event.

22.5 Request a citizen's GPS position

Send a location-request SMS

Screen /incidents/[id] · **Needs** BulkSMS configured · **Writes** citizen_share_tokens

1. Open the incident.
2. Enter the citizen's mobile number under **Contact citizen**.
3. Submit. A single-use token is minted and an SMS sent with a link to `/share-location/<token>`.
4. When they open it and allow location access, the position is written to the incident.

SECURITY The public page authenticates by the token alone and can do nothing but post a position. Tokens expire and are purged.

CHAPTER 23

Working an Incident

23.1 Advance the status

Change an incident status

Screen /responder, /incidents/[id], or the phone · **Procedure** `update_responder_status()`

1. Open the incident.
2. Press the forward button. It always names the next status in the ladder.
3. Press it at the moment the thing happens.

RECORDED The status, `unit_status`, and — on first arrival only — `arrived_timestamp`. A timeline entry is written. From the phone with no signal, the time recorded is when you pressed it, and the timeline says the update was delivered late.

23.2 Add a timeline update

Add an update

Screen /incidents/[id] or the phone · **Writes** incident_updates

1. Type into the update box.
2. Send.

APPEARS LIVE on every open copy of that incident, via realtime.

23.3 Upload evidence

Attach evidence to an incident

Screen /responder or the phone · **Writes** incident_attachments + `incident-media` bucket

1. Open the incident.
2. Choose the file and add a **caption**.
3. Upload.

STORAGE PATH `<silo>/<incident>/<kind>/…`. Attachments are embedded in the case-report PDF, so they are part of the case file rather than a side note.

23.4 Close a case

Close an incident

Who Case manager tier and supervisors · **Procedure** `close_case()`

1. Read the timeline end to end.
2. Confirm the departmental report exists where required, and that attachments are present.
3. Enter the **reason**.
4. Close.

AFTERWARDS Status *Closed*, `closed_at` stamped, the reason in the timeline, and an audit row. Not reversible through the interface.

23.5 Produce the case-report PDF

Download the case report

Endpoint /api/pdf/incident/[id] · **Rate limit** 20/min

1. Open the incident and choose the PDF download.
2. Allow it time — each scene photograph is fetched and embedded.

CONTENTS Detail, timeline, embedded attachments, audit trail — complete as one document.

CHAPTER 24

Start of Shift

24.1 The three-step gate

Complete start of shift

Screens /shift/mental-check → /shift/vehicle-selection → /shift/vehicle-checklist, or the phone Shift tab · **Resets daily**

1. **Mental readiness.** Six declarations: **memory recall**, **judgment**, **situational awareness**, **not under influence**, **anger check**, **reflexes**. Add **comments** where anything is borderline.
2. **Vehicle selection.** Choose an appliance from those available in your department. `assign_vehicle()` takes it out of the pool and attaches it to you.
3. **Vehicle inspection.** Record **fuel level** — ¼, ½, ¾ or Full — and confirm **lights functional**, **tyres condition OK**, **equipment stocked**, **comms working**, **brakes responsive**.

UNTIL ALL THREE EXIST FOR TODAY the responder app routes you to the Shift tab and you cannot open incidents. Profile stays reachable so you can always see who you are signed in as and sign out.

24.2 Going on and off duty

Go on duty

Screen phone Profile tab · **Procedure** `toggle_duty()`

1. Open **Profile** and switch **Duty status** on.
2. Allow location access when asked.

AFTERWARDS A GPS heartbeat runs every 60 seconds so dispatch can see your position. It goes through the offline queue, so losing signal does not lose your position — the newest fix is delivered when you reconnect.

End of shift

Procedure `cleanup_on_logout()`

1. Check the **sync tray** is empty. If anything is waiting and you have signal, give it a moment.
2. Sign out from **Profile**.

AFTERWARDS Your vehicle is released, duty is cleared, and your push registration is removed so the next person carrying the handset does not receive your alerts.

24.3 Unit compliance

Check unit compliance

Screen /shift/unit-compliance

1. Open **Shift → Unit Compliance**.
2. Review which units have a current inspection and which do not.

USE FOR A supervisor confirming the fleet going out is roadworthy on paper as well as in fact.

CHAPTER 25

Departmental Reports

> Each department captures its own report against the shared incident. Full field lists appear in the role playbooks (Chapters 14–16); this chapter covers what is common and what to watch.

25.1 Police case report

Screen `/police/reports/new` · **Writes** `metro_reports`. Seven capture areas: heading, officer, people, outcome, property and evidence, narrative. See 14.2 for the field-by-field procedure.

> **THE NARRATIVE IS THE REPORT**
> Every other field is structured data for searching and statistics. The **officer narrative** is what gets read in court. Write it as an account, not as notes.

25.2 Fire incident report

Screen `/fire/reports/new` · **Writes** `firefighter_reports`. Ten capture areas including timings, property, fire behaviour, response, casualties, hazmat and outcome. See 15.2.

> **TIMINGS FEED THE MEASURES**
> Dispatch, arrival, clear and total are recorded in minutes and are what the fire response-time measures are computed from. Estimate honestly rather than leaving them at zero.

PDF: download from the report page — `/api/pdf/fire-report/[id]`.

25.3 Patient care record

Screen `/medical/pcr/new` · **Writes** `paramedic_reports` plus `medical_histories`, `assessment_findings`, `vital_signs`, `interventions`, `legal_handovers`. See 16.2.

> **SIGNATURES**
> Responder, hospital representative and doctor sign on the canvas. The handover is a legal transfer of care; an unsigned PCR is an incomplete record of who was responsible for the patient and when.

25.4 Security and firearms

SIRA guard reports at `/police/security/new` with a **Resolve** flow requiring a resolution summary; firearm authorizations at `/police/firearms/new` with a **Return** action. Both are registers — the value is in them being closed out, not just opened.

25.5 Where reports are read

SCREEN	SHOWS
`/reports`	The departmental archive.
`/report-hub`	Filtered incident reporting across departments, with CSV export and saved templates.
`/reports/builder`	Custom reports: choose a dataset, filter, group and chart.

CHAPTER 26

People

26.1 Rosters and shifts

Create a shift

Screen /roster · **Who** Supervisory tier · **Writes** shifts

1. Open **Roster**.
2. Enter **shift name**, **activity type**, **start time**, **end time**, **description** and **department**.
3. Save.

GENERATORS The 224 and 24-48 rotation generators produce a pattern of shifts rather than one at a time.

Add a roster entry

Screen /roster · **Writes** rosters

1. Choose the member and the shift.
2. Set **station**, **activity type**, **call sign**, **vehicle**, **sector** and **special instructions**.
3. Save.

Record an endorsement

Screen /roster · **Writes** endorsements

1. Choose the member.
2. Set the **type** — court attendance, special event, instruction or restriction — with **details** and a **start** and **end date**.
3. Save.

WHERE IT SHOWS On the member's schedule, on the duty sheet, and in My Schedule on the phone.

26.2 Duty allocation

Fill and publish a duty day

Screen /roster/allocation · **Writes** rosters, duty_posts

1. Open **Roster → Allocation**. The board lists every **post** the shift must fill, with its minimum staffing, and shows the gaps.
2. **Assign** a member to a post: set badge, station, activity, vehicle, sector, call sign and instructions.
3. Repeat until covered. **Unassign** to correct a mistake.
4. **Publish day.** Drafts are invisible to members until this.
5. **Copy forward** to seed another day from this one.

THEN Members **ACKNOWLEDGE** their allocation from My Schedule. The engine reminds anyone who has not, 12 hours before the start.

Maintain the establishment

Screen /roster/allocation · **Writes** duty_posts

1. Use **Save post** to create or edit a post: **name**, **minimum staff**, **station**, **sector**, **vehicle**, **call sign**, **activity** and **sort order**.
2. Use **Retire post** when a post no longer exists. Retiring rather than deleting preserves history.

Print the duty sheet

Screen /roster/duty-sheet

1. Open **Roster → Duty Sheet** and choose the day.
2. Print. The layout carries shift blocks, the on-leave list, special events and sign-off columns.

26.3 Leave

Apply for leave

Screen /leave, or the phone · **Who** any operational role and Staff · **Writes** leave_entries

1. Check your **balance** for the leave type first — it is shown above the form.
2. Choose the **type** from the configured list.
3. Set **from** and **to** dates.
4. Add a **reason**. Optional, but it is what your supervisor decides on.
5. Submit.

AFTERWARDS Status *Pending*. You may only ever file your own request, and only as Pending — the decision is the supervisor's. You are alerted when they decide.

Decide a leave request

Screen /hr or /leave · **Who** Supervisory tier

1. Open **HR**; pending requests carry an inline decision.
2. Read the **coverage impact** — the share of the department away if granted.
3. Approve or reject.

AFTERWARDS `approved_by` and `approved_at` are set, the member is alerted, and the decision is audited.

26.4 Training and certifications

See 18.3. The critical field is the certification **expiry date**, which drives automatic warnings at 30, 7 and 1 days to both the member and the department.

26.5 Performance

Run the appraisal cycle

Screens /performance/kpis · /performance/plans · **Who** Supervisor and admin tier

1. **Add KPIs** for the **cycle year**: **department**, **title**, **description**, **weight**.
2. **Generate plans** for the cycle year — one per member.
3. Open a plan and **save a score** per KPI: **stage** (mid-year or final), **score**, **comments**, **evidence**.
4. **Approve the stage** to lock it.

RETIRING A KPI Use **DEACTIVATE** rather than deleting, so historical scores remain interpretable.

Read performance

Screens /performance/operational · /mine · /rollup · /reports · /role-kpis

1. **Operational** — call-taker and dispatcher throughput, category pivot, call detail, logons, with CSV export.
2. **Mine** — your own throughput against the team, monthly trend, appraisal snapshot.
3. **Rollup** — the same measures aggregated up the reporting hierarchy.
4. **Reports** — one date window by member, by department and by incident type, each with CSV.

CHAPTER 27

Reporting and Analysis

27.1 Report Hub

Build and save a filtered incident report

Screen /report-hub · **Writes** command_records (ReportTemplate)

1. Set the filters: free-text **search**, **status**, **department**, **category**, **region**, and a **from/to** date window.
2. Review the result.
3. **Export CSV**, or **save as a template** with a name so the same cut can be repeated.

27.2 Report Builder

Build a custom report

Screen /reports/builder · **Writes** report_definitions

1. Choose a **dataset** from the registry. Only datasets your role may query are offered.
2. Add **filters**. Fields, operators and enum values are validated against the registry.
3. Choose **grouping** and a **chart** type.
4. **Run**.
5. **Save** with a name, description and **visibility**, or **export CSV**.

WHY FILTERS GET DROPPED SILENTLY A filter naming a column that is not in the registry, an operator the field type does not support, or an enum value that does not exist is discarded rather than passed through to the database. This is the defence that keeps a user-composed query from becoming a way to read something it should not.

27.3 Analytics and hierarchy

SCREEN	SHOWS
`/analytics`	Trends, closure rate, volume charts. Admin, Viewer and Supervisor.
`/hierarchy`	The reporting structure, with CSV export.
`/hierarchy/person`	Set an individual's reporting line.
`/dispatch/kpi`	Dispatcher throughput and lifecycle timings.

27.4 Search

Find a record from any string

Screens ⌘K anywhere, or /search · Procedure `search_everything()`

1. Press ⌘K / `Ctrl+K`, or open **Search**.
2. Type at least two characters. Partial identifiers work — part of an incident number, a plate, a badge, a surname.
3. Results group by kind: incidents, personnel, fleet, police reports, fire reports, security reports.

SCOPE Results are RLS-scoped, so you see only what you could open anyway. The `/search` page keeps the query in the URL, so a search can be bookmarked or sent to a colleague.

CHAPTER 28

Surveillance and External Systems

28.1 The camera wall

View cameras

Screen /cameras · **Endpoint** /api/cctv/cameras

1. Open **Cameras**. The registry served is SANRAL today, or the live HikCentral estate once that integration is enabled.
2. Select a camera for live preview.

28.2 Retrieving footage

Retrieve recorded footage

Screen /cameras/footage · **Endpoint** /api/cctv/stream · **Audited**

1. Open **Cameras → Footage**.
2. Choose the camera and a time window — capped at 24 hours per request.
3. Request the playback URL.

CHAIN OF CUSTODY Every grant is written to the audit log with your name. Footage retrieval is evidence handling, and who requested what has to be answerable later.

28.3 Radio (PTT)

Send text to a radio

Screen /dispatch/ptt · **Who** Dispatch roles · **Rate limit** 60/min

1. Open **Dispatch → PTT**.
2. Enter the destination **account** and the **message**.
3. Send. Your name is derived from your profile and attached as the sender.
4. The two-hour audio log is on the same screen.

28.4 External feeds

FEED	WHERE IT APPEARS	NEEDS
Traffic incidents	Cameras, dispatch console	`TOMTOM_API_KEY`
Points of interest	Map layers	`TOMTOM_API_KEY`
Aircraft	Disaster monitor	`FLIGHTRADAR_API_KEY`
Vessels	Disaster monitor	`MARINESIA_API_KEY`
Wind and weather	Disaster monitor	`OPENWEATHER_API_KEY`
Camera registry import	Cameras	`ITRAFFIC_API_KEY`

> **WHEN A FEED IS NOT CONFIGURED**
> The panel says so explicitly rather than showing an error or, worse, plausible-looking data. A control room must be able to tell "this feed is off" from "this feed says there is nothing there".

VOLUME VI

The Responder App

The phone a crew carries on shift, designed around three failures that will happen: no signal, a locked screen, and no free hand.

CHAPTER 29

Screens and the Duty Gate

29.1 The screens

SCREEN	PURPOSE
Incidents (tab)	Assigned calls, severity first. A **Mine / Nearby** switch shows every live call in the department sorted by distance — read-only.
Alerts (tab)	Inbox with a live unread badge and acknowledgement state.
Schedule (tab)	Current duty, next duty, upcoming allocations with call sign, vehicle and sector, leave balances, endorsements.
Shift (tab)	The three-step start-of-duty wizard and unit compliance.
Profile (tab)	Identity, duty toggle, push diagnostics, delivery status, self-service links, sign-out.
Incident detail	Full call, navigation handoff, caller dial, status advance, evidence capture, timeline.
Leave	Apply, with balances shown before the form.
Criti	The assistant, scoped to your own calls and shift.

Two overlays sit above every screen: the **SOS button** (Chapter 32) and the **sync tray** (Chapter 31).

29.2 The duty gate

You cannot open incidents until today's readiness flow is complete. On opening the app with it outstanding, you are routed to the Shift tab.

> **TWO DELIBERATE ESCAPES**
>
> **Profile stays reachable** — somebody who fails the readiness check can never satisfy the gate, which is correct, but must still be able to see who they are signed in as and sign out. A gate with no exit is a trap, not a control. And if the check **cannot be evaluated** because the device is offline, you are let through: the server remains the record of truth, and stranding a crew behind a check we cannot run helps nobody.

CHAPTER 30

Mobile Task Procedures

30.1 Sign in

Screen Login

1. Enter your Criticom email and password — the same credentials as the console.
2. The session persists, so you will not normally sign in again during a tour.

IF IT HANGS ON "RESTORING YOUR SESSION" the app has failed through to login rather than trapping you. Sign in again.

30.2 Work an assigned call

Screen Incidents tab → detail

1. Open the call from the Incidents tab. Critical calls carry a red strip: *Priority 1 — respond immediately*.
2. **Navigate** hands the coordinates to the phone's own map application for turn-by-turn.
3. **Call caller** dials the number on the incident.
4. Advance the status with the single forward button as each thing happens.
5. Add timeline updates and evidence while on scene.

30.3 See what else is live nearby

Screen Incidents tab → Nearby

1. Switch to **Nearby**.
2. Allow location access to sort by distance; without it you still get the list, unsorted.

READ-ONLY You cannot advance the status of a call assigned to another crew. Situational awareness, not a second dispatch desk.

30.4 Capture evidence

Screen incident detail → Scene evidence

1. **Camera** — take a photograph. **Library** — attach an existing one.
2. **Record voice note** — tap to start; the control turns red and counts. Tap again to stop and file it.

COVERAGE IS IRRELEVANT Files are copied out of the OS cache into storage the app controls, then uploaded when there is signal. A photograph of a scene that has since been cleared cannot be retaken, which is why it is retained rather than left in the cache.

30.5 Acknowledge an alert

Screen Alerts tab, or the full-screen interrupt

1. A High or Critical dispatch opens a full-screen alert. Choose **Acknowledge & open** to go straight to the call, or **Acknowledge** to clear it.
2. Anything you missed is in the **Alerts** tab. Opening an unread alert there acknowledges it if acknowledgement was required.
3. **Mark all read** clears the badge.

THE BANNER AT THE TOP OF THE INBOX counts alerts still waiting on your acknowledgement.

30.6 Apply for leave

Screen Profile → Apply for leave

1. Check your balances at the top.
2. Choose a type, set the dates as `YYYY-MM-DD`, add a reason.
3. Submit. The summary line tells you how many days and what would remain — and warns if it exceeds your balance.

EXCEEDING YOUR BALANCE DOES NOT BLOCK SUBMISSION. Your supervisor decides.

30.7 Ask Criti

Screen Profile → Ask Criti

1. Ask about your own calls, your crew or your shift.
2. Suggested questions are offered on the empty screen.

SCOPE Criti sees exactly what you can see. If a question fails, the text is put back in the box so you need not retype it one-handed.

CHAPTER 31

The Offline Outbox

> Every field action is written to the phone before it is sent. This is why a dead zone costs you nothing, and why you should never press a button twice.

31.1 What you see

INDICATOR	MEANING
Waiting to send on a card	Your change is recorded on the phone and will deliver itself.
Sync pill above the tab bar	Count of undelivered items. Amber when offline, red when something failed.
Sync tray (tap the pill)	Every undelivered item, when it was captured, how many attempts, and the error if it failed.

31.2 What to do

1. **Nothing, normally.** The queue delivers on reconnect, on returning to the app, and on a timer.
2. **Before signing out**, check the tray is empty. Work is held against *your* account — if you sign out with items queued they wait for you, not for the next person carrying the handset.
3. **If something shows as failed**, open the tray and read the error. **Try again now** retries everything.
4. **Discard** only if you are certain the item is no longer needed. It is deleted and never reaches the case file.

31.3 What the queue guarantees

GUARANTEE	CONSEQUENCE FOR YOU
Honest times	The recorded time is when you pressed the button, not when signal returned.
Nothing dropped	An item that cannot be delivered is parked and stays visible until a human deals with it.
Safe replay	If delivery is interrupted mid-flight, the retry cannot create a duplicate.
Emergency first	A panic is delivered before a queue of photographs.
Your work only	Items are delivered under the account that captured them.

> **RETRIES ONLY COUNT WHEN YOU HAVE SIGNAL**
> The queue refuses to attempt delivery while the device is offline, so the retry budget is not spent driving through a tunnel. Eight failures means the server is rejecting the write — a real problem worth reporting.

CHAPTER 32

Officer-Down

Call for emergency assistance

Control the SOS button, on every screen · **Procedure** `raise_panic()`

1. **Press and hold** the SOS button. It reads **HOLD…** and fills.
2. The phone taps you at each third of the hold, so you can feel it arming without looking.
3. After 2.5 seconds it fires. Release earlier and nothing happens.
4. A confirmation tells you whether your position was captured.

WHAT HAPPENS A Critical incident is created at your position and a blocking alert is raised to your own department *and* the control room. With no signal it is queued at the highest priority and sent the moment you have any. Position is raced against a four-second timeout — if no fix is available it sends anyway, marked position unknown.

> **USE THE RADIO TOO**
> The platform is a second channel, not a replacement. Raise it on the radio as well if you can.

If you trigger it accidentally: say so on the radio, and ask dispatch to close the incident with a note. There is no cancel — the button is deliberately easy to trigger and impossible to un-send, because hesitation is the worse failure.

CHAPTER 33

Alerting and Push

33.1 Checking a handset before a shift

Open **Profile → Dispatch alerts**. It states exactly one of:

MESSAGE	MEANING AND ACTION
Push enabled	The handset will ring for a dispatch even locked. Nothing to do.
Notifications blocked	Permission was declined. **Fixable here** — turn notifications on for Criticom in device settings.
Push not in this build	This build has no push service configured. **Not fixable on the device** — report it; the handset should not go out on shift.
Push unavailable on this device	An emulator or simulator. Expected in testing.
Push registration failed	Something went wrong registering. Report it for checking.

In every one of these cases, alerts still arrive instantly *while the app is open*.

33.2 Why there are several channels

Realtime covers open apps; push covers locked phones; a local notification covers an alert arriving while the app is backgrounded; the inbox covers everything afterwards. Each alone has a blind spot, and a missed dispatch is not recoverable — so they overlap deliberately, and duplicates are deduplicated by alert id.

VOLUME VII

Administration

Everything under `/admin`: who exists, what the service calls things, how hard the platform pushes, and which external systems are switched on.

CHAPTER 34

Users, Roles and Personnel

34.1 Set a user's role

Screen /admin · **Writes** profiles, audit_logs

1. Open **Admin → Users & Roles** and filter by department if needed.
2. Find the person and choose their **role** from the 23 assignable values.
3. Save.

SET THE SILO TOO. A role only opens a departmental section when the user's silo matches — `Paramedic` in the MetroPolice silo opens nothing under `/medical`.

34.2 Add a personnel record

Screen /admin/personnel · **Writes** personnel

1. Enter **silo**, **first name**, **last name**, **badge number** and **rank**.
2. Save.

PERSONNEL AND USERS ARE SEPARATE. A personnel record is the operational person; a user is a login. Linking them by `user_id` is what lets the phone find "my" incidents and schedule.

34.3 Add a control-room operator

Screen /admin/operators · **Writes** system_profiles

1. Enter **first name** and **last name**.
2. Save. A call sign is generated.

34.4 Add or retire an appliance

Screen /admin/vehicles · **Writes** apparatus

1. To add: enter **silo**, **unit number**, **name**, **type**, **make**, **model**, **licence plate** and **station number**.
2. To take one out of service: set its **status** — Out of Service, Maintenance or Decommissioned.

STATUS IS OPERATIONAL. Only *Available* appliances are offered at vehicle selection and on the assignment console.

34.5 Register a device

Screen /admin/devices · **Writes** devices

1. Enter **silo**, **device ID**, **device type**, **model** and **assigned station**.

CHAPTER 35

Configuration

> Admin → Configuration holds the lists the service can change without a deployment. Everything here is soft-deleted — toggled inactive rather than removed — so historical records stay interpretable.

35.1 Add a configuration record

Screen /admin/config · **Writes** command_records

1. Choose the **module**: Incident Category, Leave Type, Shift Type, Report Template or Ops Link.
2. Enter the **name** and **code**.
3. For an incident sub-category, set the **parent** group; leaving it blank makes this row a group.
4. For a category, add extra form **prompts**, one per line, to appear on the log-call screen.
5. Save.

LEAVE TYPES CARRY ENTITLEMENTS. The annual day allowance lives on the leave-type record, which is what the balances on the leave screens are computed against.

35.2 Repoint an external ops link

Screen /admin/config, module OpsLink

1. Find the link — E-NATIS, CAS, PERSAL, FIREARM, FINES, TRACE, FORENSICS and the rest.
2. Set its endpoint.

UNTIL AN ENDPOINT IS ISSUED the link renders dimmed in the command bar rather than looking broken.

35.3 Retire a configuration record

Screen /admin/config

1. Toggle the record inactive.

IT STOPS BEING OFFERED on capture forms but existing records that reference it still read correctly.

CHAPTER 36

Operations Settings

> Twelve knobs at `/admin/operations`. Each declares a type, bounds and a default; values are clamped on read and write, every change is audited, and if the table is unreachable the code defaults apply — a tuning problem can never take a screen down.

SETTING	EFFECT
`sla.clock_warn_seconds`	When the intake clock on the log-call screen turns amber.
`sla.clock_critical_seconds`	When it turns red.
`sla.response_target_minutes`	The response target the dashboard gauge measures against.
`sla.call_duration_target_minutes`	The call-handling target.
`alerts.modal_min_severity`	**Which severity interrupts with a blocking pop-up.** The single most operationally significant knob here.
`alerts.sound_enabled`	Whether an alert plays a tone.
`alerts.toast_seconds`	How long a corner toast stays.
`dispatch.eta_speed_kmh`	Assumed speed in the ETA model.
`dispatch.eta_road_factor`	Road-distance multiplier over straight line.
`map.default_basemap`	Which basemap loads first.
`board.feed_items`	How many items the ops board feed shows.
`board.volume_days`	The window the volume chart covers.

Change a setting

Screen /admin/operations

1. Edit the value and save. If it is out of bounds it is clamped, and the field updates to show what was actually stored.
2. **Reset** removes the override and returns to the built-in default.

> **RAISING `ALERTS.MODAL_MIN_SEVERITY` IS AN OPERATIONAL DECISION**
>
> Set it to Critical and High-severity dispatches stop interrupting crews. That may be right for a busy department, but it changes what people are pulled away from other work for. Record why.

CHAPTER 37

Automations

37.1 What runs

Seven rules in the database on a five-minute tick. Everything Criticom did before wave 5 happened because a person did it — right for dispatching, wrong for noticing that a call has been waiting twenty minutes or a licence lapses next week.

37.2 The rules

RULE	EVERY	BEHAVIOUR AND DEFAULTS
Unassigned call escalation	5 min	High alert at 10 minutes waiting; Critical blocking interrupt at 20. To the owning department and the control room.
No arrival recorded	10 min	Dispatched or En Route with no arrival after 30 minutes.
Open incident gone quiet	30 min	Active incident with no timeline entry for 4 hours. At most one nudge per incident per day.
Certification expiry	12 h	Warns at 30, 7 and 1 days — to the member and the department.
Unacknowledged duty	1 h	Published duty starting within 12 hours that the member has not confirmed.
Morning digest	30 min	Once per department per day at the configured hour: volume, criticals, closure rate, average response.
Housekeeping	24 h / 6 h	Prunes expired alerts, old run logs, spent suppression records and stale rate-limit counters.

37.3 Managing them

Tune, disable or run a rule

Screen /admin/automations · Who Admin tier · Audited

1. **Read the health banner first.** It says whether the engine is ticking, when the last tick was, and how many rules are erroring.
2. To retime: edit the rule's thresholds and its **check every** interval, then **Save**. Out-of-range values are clamped and the field shows what was stored.
3. **Run now** executes the rule immediately, ignoring its interval, and reports what it did.
4. **Disable** stops it. Use this when a rule is misbehaving — a disabled rule is a known gap; a silently failing one is not.

RECENT RUNS are listed below the rules with what each did and how many things it acted on.

37.4 If the engine is not running

The status page and the automations page both say so explicitly. The engine is complete without `pg_cron` — anything with a clock can drive it:

```
select public.run_automations();
```

Point an external scheduler at that every five minutes and the service is fully functional while cron is fixed.

> **WHY YOU WILL NOT GET SPAMMED**
> Every action is claimed against a unique `(rule, entity, tier)` record, so a rule running every five minutes alerts once per escalation tier rather than every five minutes forever. If you see the same incident escalate twice, it has crossed from warn to critical — a real deterioration.

CHAPTER 38

Integrations

38.1 The catalogue

CATEGORY	INTEGRATION	PURPOSE
Core	Supabase	Database, auth, storage, realtime.
	Esri World Imagery	Map basemap.
AI	Groq	Primary provider behind Criti.
	Google Gemini	Fallback provider.
Comms	RealPTT	Text to radio, audio log.
	Zoiper	Softphone dial-out.
Notifications	Expo Push	Responder app notifications.
	BulkSMS	Citizen SMS and GPS requests.
Surveillance	SANRAL CCTV Matrix	The current freeway camera wall.
	Hikvision HikCentral	The incoming CCTV platform.
External feeds	TomTom	Traffic incidents and points of interest.
	FlightRadar24	Aircraft positions and tracks.
	AIS / Kpler	Vessel positions, Durban harbour.
	OpenWeather	Wind for fire-spread modelling.
	i-Traffic	Camera registry import.

38.2 Enabling one

Enable and test an integration

Screen /admin/integrations

1. Set the integration's configuration.
2. Ensure the corresponding environment variable is set in the hosting environment — **and redeploy**, because these are read at build time.
3. Toggle it **enabled**.
4. Run the **connection probe** on its row.

KEYS NEVER REACH THE BROWSER. Every third-party call goes through a server route that holds the key, requires a session and is rate limited.

38.3 The CCTV cut-over

`/api/cctv/cameras` serves the SANRAL registry until Hikvision is enabled with an app key, secret and base URL — then the same endpoint serves the live HikCentral estate. The cut-over is a toggle in Admin, not a deployment. Roll back the same way.

CHAPTER 39

Branding

Set a department's identity

Screen /admin/departments · **Writes** department_branding

1. Choose the department.
2. Set **official name**, **short name**, **monogram**, **motto**, **primary colour** and **accent colour**.
3. Upload the crest, or use **Clear logo** to remove it.

WHY CRESTS ARE UPLOADED, NEVER SHIPPED Official municipal and service insignia belong to the municipality. Until one is uploaded the interface renders a monogram badge in the department's colours, so nothing looks unfinished. The identity appears in the sidebar, My Schedule, and the mobile shell and profile.

VOLUME VIII

Running It

Security, deployment, recovery and what to do at three in the morning.

CHAPTER 40

Security

40.1 The model

LAYER	CONTROL
Transport	HTTPS throughout; HSTS with a two-year max-age and preload.
Session	Supabase Auth. Role and silo in JWT app metadata, which the client cannot alter.
Route	Middleware evaluates the route matrix; unauthorized → `/403`.
Row	RLS on every table. Deny by default, then explicit policies.
Function	`require_role()` opens every privileged mutator; `EXECUTE` revoked from `anon`.
Rate	Per-user limits on routes that cost money or third-party quota.
Audit	`audit_logs` on every mutation with actor, department and detail.

40.2 Headers

HEADER	VALUE / NOTE
`Content-Security-Policy`	Self by default; Supabase REST and websockets, map tile hosts and the key-free geocoding upstreams allow-listed. `frame-ancestors 'none'`.
`Strict-Transport-Security`	`max-age=63072000; includeSubDomains; preload`
`X-Frame-Options`	`DENY`
`X-Content-Type-Options`	`nosniff`
`Referrer-Policy`	`strict-origin-when-cross-origin`
`Permissions-Policy`	Camera, microphone, geolocation to self — the console uses all three. Everything else denied.
`Cache-Control`	`no-store` on every API response.

> **KNOWN LOOSENING**
>
> `script-src` permits `'unsafe-inline'` and `'unsafe-eval'` for Next's inline bootstrap; `style-src` permits `'unsafe-inline'` for Tailwind's runtime theme variables and per-department colours. Tightening needs per-request nonces — separate work.

40.3 Rate limiting

BUCKET	LIMIT	RATIONALE
`chat`	40 / min	A person types perhaps 6 a minute; a render loop types 600.
`geocode`	120 / min	Fires per keystroke-batch on the address field.
`external`	90 / min	Metered upstreams; panels poll on a timer.
`pdf`	20 / min	Embedding photographs is the most expensive request served.
`ptt`	60 / min	Radio traffic is bursty.
`cctv`	60 / min	Each grant is audit-logged; a flood pollutes the trail.

Counted in Postgres because the console deploys serverless — an in-memory counter is per-instance, and ten concurrent functions would mean ten times the limit. Identity comes from `auth.uid()` inside the database function, so a handler cannot get it wrong and a caller cannot spend somebody else's budget.

> **IT FAILS OPEN**
> If the limiter errors, the request proceeds. A rate limiter that takes down dispatch when the database hiccups causes more harm than the runaway loop it guards against.

If an operator hits a 429 in normal work, the limit is wrong, not the operator. Raise it in `apps/web/lib/rate-limit.ts` and deploy. To clear one person's counter:

```
delete from rate_limit_buckets where key like 'chat:%<user-uuid>';
```

40.4 Outstanding items

> **OPEN, AND TO BE CLOSED BEFORE OR AT HANDOVER**
>
> - **V4 leaked credentials.** The previous system's BulkSMS token pair and Gemini key are in that repository's source history, including a hardcoded token in its notifications controller. They must be **revoked at the provider** — rotating them in V5's environment does not help, because the old values remain in V4's history.
> - **Setup-time exposure.** The Supabase service-role key and database password were pasted into a chat transcript during provisioning and should be rotated.
> - **CSP nonces.** Removing `'unsafe-inline'` and `'unsafe-eval'`.
> - **Accessibility.** 19 tracked warnings, mostly labels not programmatically associated with inputs on the call-taking screen — a screen reader announces those fields with no name.

CHAPTER 41

Deploying

41.1 Console

Push to `main` ; CI must be green; Vercel builds the same commit. Environment changes need a **redeploy**, not a restart — `NEXT_PUBLIC_*` values are compiled into the bundle.

41.2 Database

```
set -a && . ./.env && set +a
node scripts/apply-migration.mjs supabase/migrations/00NN_your_migration.sql
```

- Migrations are append-only and numbered. Never edit an applied one — write the next.
- Each file runs in a transaction, so a failure rolls the whole file back.
- Apply **before** the deployment that depends on them: the app tolerates a schema newer than the code far better than the reverse.

> **POSTGREST RESOLVES RPCS BY ARGUMENT NAME**
> Adding a parameter with a default to an existing function creates a *second* overload, and a call matching both resolves to neither. Drop the old arity in the same migration.

41.3 Responder app

```
eas build  --profile production --platform android   # native binary
eas build  --profile preview    --platform android   # internal APK, no push
eas update --branch  production --message "what changed"   # JavaScript only
```

> **THE OTA TRAP — READ BEFORE PUBLISHING AN UPDATE**
> An over-the-air update ships **JavaScript only**. If the new JavaScript imports a native module the installed binary does not contain, every handset crashes at launch. The defence is `runtimeVersion` , which follows the app version: **bump the version whenever you add a native module**, and older binaries become ineligible for the update instead of being handed something that cannot boot. Wave 5 added five native modules and moved the app to 1.1.0 for exactly this reason.

The Android FCM contract

A production Android build **refuses to build** without `google-services.json` , rather than producing an app that cannot receive a dispatch. Supply it as an EAS secret:

```
eas secret:create --scope project --name GOOGLE_SERVICES_JSON \
  --type file --value ./google-services.json
```

The `preview` profile carries `CRITICOM_ALLOW_NO_PUSH=1` for internal builds without push. Those are fine for testing and **should not go out on shift** — alerts arrive only while the app is open.

CHAPTER 42

Backup and Recovery

Supabase takes automatic daily backups; point-in-time recovery depends on the plan. **Verify which you have before you need it.**

```
pg_dump "$SUPABASE_DB_URL" --no-owner --no-acl -Fc -f criticom-$(date +%F).dump
```

Restore into a *new* project first and check it, then cut over. Restoring over a live database while crews are working turns one incident into two.

NOT COVERED BY A DATABASE RESTORE	CONSEQUENCE
Storage buckets `incident-media`, `avatars`, `department-logos`	Evidence photographs live here. A restore that ignores storage silently loses the attachments a case file references.
Accounts created since the snapshot	Re-create with `pnpm db:create-admin`.
Push tokens	Harmless — phones re-register on next launch.

CHAPTER 43

First Response

BEFORE ANYTHING ELSE

Every procedure here is slower than the radio. If the platform is degraded and crews are waiting, put the control room on radio and paper, then fix the platform.

43.1 Triage order

#	CHECK	INTERPRETATION
1	**Admin → System Status**	Each check states what it means. Start here.
2	`/api/health/config`	Which environment variables the running build can see. Missing means absent at *build* time — redeploy.
3	`psql "$SUPABASE_DB_URL" -c 'select now()'`	If this fails, everything else is downstream.
4	Sign in as a known account	Sign-in failing while the database answers points at Auth, not data.
5	Alerts in the last 15 minutes	Zero on a busy shift means realtime or the automation engine.

43.2 Common situations

SITUATION	RESPONSE
A phone is not receiving dispatches	Profile → Dispatch alerts states which of five cases applies (33.1). Realtime still delivers while the app is open in all of them.
"My update did not go through"	The sync tray lists everything undelivered with its error. Queued work is held against that person's account and sends when they next have signal while signed in.
A panic fired by accident	Close the incident with a note. It is deliberately impossible to un-send.
Alert storm from a rule	Disable the rule from Admin → Automations, then inspect `automation_events` — repeat alerting means its claim key is wrong. Widening a threshold is the fast mitigation.
Automations not running	See 37.4. Drive `run_automations()` from any external scheduler while cron is fixed.
Storage filling up	Scene photographs dominate. Nothing deletes them automatically, deliberately — they are evidence. Removing any is a records-retention decision.
Incident numbers look odd	Format is `INC-YYYYMMDD-NNNN`, sequential per department per day. Numbers before migration 0024 used a random suffix and are not sequential — history, not a fault.

43.3 Useful queries

```
-- Live officer emergencies
select incident_number, silo, responding_officer, location_address, incident_created_at
  from incidents
 where source = 'Responder-Panic' and status not in ('Closed','Resolved');

-- Calls waiting for a department right now
select incident_number, silo, incident_severity,
       round(extract(epoch from (now() - call_timestamp)) / 60) as minutes_waiting
  from incidents
 where status in ('New','Pending','Pending Dispatch','Pending Analysis')
 order by call_timestamp;

-- Automation health
select key, enabled, last_status, last_run_at, last_detail
  from automation_rules order by key;

-- Recent sign-ins
select user_name, silo, timestamp from audit_logs
 where action_type = 'Logon' order by timestamp desc limit 20;
```

CHAPTER 44

Engineering Standards

44.1 CI

STEP	CATCHES
`pnpm lint`	Defect-class lint rules.
`pnpm typecheck`	Four projects: db, api, web, mobile.
`pnpm test`	47 tests across three packages.
`pnpm build`	The production console build.
Migrations from empty	All 28 applied in order to a throwaway PostGIS database.

WHY THE MIGRATION JOB MATTERS

A migration is the one artefact with no undo, and the sequence from nothing is what a new environment and a restore both execute. A migration that only works against the current production database has not been tested.

44.2 Testing

`node --test` with `--experimental-strip-types`. No framework, no transpile, no config — Node 22 runs the TypeScript directly. Imports use explicit `.ts` extensions.

SUITE	PINS
`roles.test.ts`	The route matrix, including the two-desk split.
`reporting/query.test.ts`	That the report builder drops unknown columns, unsupported operators and invalid enum values.
`automations.test.ts`	That a zero threshold can never be stored.
`outbox-policy.test.ts`	Queue ordering, collapsing, retry classification, ownership.

A PATTERN WORTH COPYING

When safety-relevant logic is tangled with an untestable runtime, split the pure decisions into their own module and test that. `outbox-policy.ts` exists for this; `outbox.ts` re-exports it so no call site changed. Writing those tests immediately found a real defect — retry attempts were accruing while offline, so a crew in a tunnel would have a scene photograph marked failed.

44.3 Linting

`pnpm lint` runs `biome lint`, not `biome check`. The formatter disagrees with this codebase's consistent hand-formatting; running it produces a 150-file diff that changes no behaviour and buries every real finding. Rules the house style diverges from are off, each reviewed against its actual violations and recorded with count and reason in `docs/07-ENGINEERING-STANDARDS.md`.

44.4 Conventions

- Server-action forms with uncontrolled fields.
- Queries live in `packages/api/src/<module>`; pages never build queries inline.
- `logAudit` on every mutation.
- Prefer `SECURITY INVOKER` and let RLS scope; reach for `DEFINER` only where atomicity or cross-department writes demand it, and guard it.
- Comments explain **why**, not what.

VOLUME IX

Reference

Indexes for looking things up: every action and its fields, routes, endpoints, tables, procedures, migrations, environment variables and terms.

APPENDIX A

Action Index

> All 67 server actions with the fields each expects. This is the definitive list of what a user can change, and it is generated from the code.

ACTION	SCREEN	FIELDS
setRoleAction	/admin	id, role
createApparatusAction	/admin/vehicles	silo, unit_number, name, type, make, model, license_plate, station_number
setApparatusStatusAction	/admin/vehicles	id, status
createDeviceAction	/admin/devices	silo, device_id, device_type, model, assigned_station
createPersonnelAction	/admin/personnel	silo, first_name, last_name, badge_number, rank
createOperatorAction	/admin/operators	first_name, last_name
addConfigAction	/admin/config	module, name, code, parent, prompts
toggleConfigAction	/admin/config	id
saveSettingAction	/admin/operations	key, value
resetSettingAction	/admin/operations	key
toggleAutomationAction	/admin/automations	key, enabled
saveAutomationAction	/admin/automations	key, interval_minutes, + rule thresholds
runAutomationAction	/admin/automations	key
setIntegrationEnabledAction	/admin/integrations	key, enabled
saveIntegrationConfigAction	/admin/integrations	key, config
saveBrandingAction	/admin/departments	silo, primary_color, accent_color, official_name, short_name, monogram, motto
clearLogoAction	/admin/departments	silo
addUpdateAction	/incidents/[id]	incident_id, message, silo
setStatusAction	/incidents/[id]	incident_id, status, silo
requestCitizenLocationAction	/incidents/[id]	incident_id, silo, phone
closeCaseAction	/incidents/[id]	incident_id, silo, reason
responderStatusAction	/responder	incident_id, status, silo
uploadEvidenceAction	/responder	incident_id, silo, caption, file
mentalCheckAction	/shift/mental-check	six declarations, comments
assignVehicleAction	/shift/vehicle-selection	apparatus_id
vehicleChecklistAction	/shift/vehicle-checklist	apparatus_id, fuel_level, five confirmations
createMetroReportAction	/police/reports/new	39 fields — see 14.2

ACTION	SCREEN	FIELDS
createFirearmAction	/police/firearms/new	firearm authorization fields
returnFirearmAction	/police/firearms	id
createSecurityReportAction	/police/security/new	guard, SIRA, firearm, GPS fields
resolveSecurityReportAction	/police/security	id, resolution_summary
createFireReportAction	/fire/reports/new	63 fields — see 15.2
publishPirAction	/fire/pir	firefighter_report_id
createPcrAction	/medical/pcr/new	patient, history, assessment, vitals, interventions, handover
updatePcrAction	/medical/pcr/[id]/edit	pcr_id + changed fields
addRestockAction	/medical/restock	unit_id, drug_name, qty_used, qty_remaining
dispatchResourcesAction	/disaster/dispatch	resources, partners
periodicUpdateAction	/disaster	log_id, text
postMortemAction	/disaster/post-mortem	post-mortem fields
createShiftAction	/roster	shift_name, activity_type, start_time, end_time, description, department
createRosterEntryAction	/roster	user_id, user_name, badge_number, shift_id, station, activity_type, description, call_sign, vehicle, sector, special_instructions
addEndorsementAction	/roster	user_id, employee_name, badge_number, type, details, start_date, end_date
deactivateEndorsementAction	/roster	id
assignAction	/roster/allocation	userId, userName, silo, postId, badge, station, activity, vehicle, sector, callSign, instructions
unassignAction	/roster/allocation	id
publishDayAction	/roster/allocation	silo
copyForwardAction	/roster/allocation	from, to, silo
savePostAction	/roster/allocation	name, minStaff, id, silo, station, sector, vehicle, callSign, activity, sortOrder
retirePostAction	/roster/allocation	id
acknowledgeAction	/roster/allocation	id
applyLeaveAction	/leave	eff_from, eff_to, rank, service_no, absence, reason
setLeaveStatusAction	/leave, /hr	id, status

ACTION	SCREEN	FIELDS
addTrainingAction	/training	personnel_id, training_name, date_completed, notes_learned
addCertificationAction	/training	personnel_id, name, expiry_date, issued_by, date_achieved, remarks
addKpiAction	/performance/kpis	cycle_year, department, title, description, weight
deactivateKpiAction	/performance/kpis	id
generatePlansAction	/performance/plans	cycle_year
saveScoreAction	/performance/plans/[id]	plan_id, kpi_id, kpi_title, stage, score, comments, evidence
approveStageAction	/performance/plans/[id]	plan_id, stage
setReportsToAction	/hierarchy/person	silo, member_kind, member_id, choice
saveTemplateAction	/report-hub	name, q, status, silo, category, region, from, to
runReportAction	/reports/builder	report spec
exportCsvAction	/reports/builder	report spec
saveReportAction	/reports/builder	name, spec, visibility, chart, id, description
deleteReportAction	/reports/builder	id
updateProfileAction	/profile	profile fields
generateCallSignAction	/profile	first_name, last_name

APPENDICES B–C

Routes and Endpoints

B — Console routes by area

AREA	ROUTES
Core	/dashboard · /notifications · /profile · /profile/my-hr · /search · /assistant · /403
Dispatch	/dispatch/log-call · /dispatch/pending · /dispatch/assign/[id] · /dispatch/kpi · /dispatch/ptt
Incidents	/incidents · /incidents/[id] · /incidents/map
Responder	/responder · /shift/mental-check · /shift/vehicle-selection · /shift/vehicle-checklist · /shift/unit-compliance · /shift/reason
Police	/police · /police/reports(/new,/[id]) · /police/firearms(/new) · /police/security(/new) · /police/crime-hotspots
Fire	/fire · /fire/reports(/new,/[id]) · /fire/hydrants · /fire/pir · /fire/pir/publish/[reportId]
Medical	/medical · /medical/pcr(/new,/[id],/[id]/edit) · /medical/restock
Disaster	/disaster · /disaster/dispatch · /disaster/partners · /disaster/post-mortem
People	/hr · /leave · /training · /roster · /roster/allocation · /roster/duty-sheet · /roster/mine · /hierarchy · /hierarchy/person
Performance	/performance · /kpis · /mine · /operational · /plans · /plans/[id] · /reports · /role-kpis · /rollup
Reporting	/reports · /reports/builder · /report-hub · /analytics · /case-manager · /case-manager/audit
Surveillance	/cameras · /cameras/footage
Admin	/admin · /config · /departments · /devices · /integrations · /operations · /operators · /personnel · /status · /vehicles · /automations
Public	/login · /share-location/[token]

C — API endpoints

ENDPOINT	PURPOSE
`/api/chat`	Criti. Cookie or bearer authentication. 40/min.
`/api/geocode`	Address ⇄ coordinate. Session-gated, 120/min.
`/api/route`	Road-routed ETA.
`/api/wind`	OpenWeather proxy.
`/api/external/{traffic,pois,flights,flight-tracks,vessels}`	Metered upstream proxies. 90/min.
`/api/cameras/import`	i-Traffic camera registry import.
`/api/cctv/cameras`	Camera registry — SANRAL or HikCentral.
`/api/cctv/stream`	Signed preview and playback URLs. Audited, 60/min.
`/api/pdf/incident/[id]`	Case-report PDF with embedded photographs. 20/min.
`/api/pdf/fire-report/[id]`	Fire report PDF. 20/min.
`/api/push/notify`	Expo push fanout to routed departments.
`/api/ptt/{send,audio}`	RealPTT text-to-radio and audio log. 60/min.
`/api/{hr/report,performance/operational,report-hub/export,hierarchy/export}.csv`	CSV exports.
`/api/integrations/test/[key]`	Connection probe. Admin only.
`/api/health/config`	Deployment diagnostics. Liveness public; detail admin only.

APPENDICES D–F

Data, Procedures, Migrations

D — Tables by domain

DOMAIN	TABLES
Identity	profiles, system_profiles, personnel, reporting_lines
Incidents	incidents, incident_updates, incident_attachments, incident_number_counters, citizen_share_tokens
Fleet	apparatus, devices, cameras, hydrants
Police	metro_reports, metro_offenses, firearm_authorizations, security_reports, crime_stats
Fire	firefighter_reports, public_incident_reports
Medical	paramedic_reports, medical_histories, assessment_findings, vital_signs, interventions, legal_handovers, ambulance_restocks
Disaster	disaster_logs, post_mortem_reports, relief_stocks, mutual_aid_partners
Shift & duty	shifts, rosters, duty_posts, endorsements, mental_checks, vehicle_checklists
HR	leave_entries, training_records, certifications
Performance	performance_kpis, performance_plans, performance_scores
Alerting	alerts, alert_reads, notifications, push_tokens
Configuration	command_records, app_settings, integration_settings, department_branding, report_definitions
Platform	audit_logs, automation_rules, automation_runs, automation_events, rate_limit_buckets

E — Key stored procedures

FUNCTION	SECURITY	PURPOSE
`log_incoming_call()`	Definer	Creates an incident, allocates its number, audits.
`next_incident_number()`	Definer	Collision-free `INC-YYYYMMDD-NNNN`.
`check_duplicate_incident()`	Definer	12-hour duplicate lookup.
`route_incident()`	Definer	Route, timeline, audit, alerts — one transaction.
`dispatch_unit()`	Definer	Commits a specific appliance.
`update_responder_status()`	Definer	Status change, accepting the time the responder acted.
`raise_panic()`	Definer	Officer-down. No role gate, by design.
`close_case()`	Definer	Closure with reason, audited.
`toggle_duty()`	Definer	Duty on/off with position.
`assign_vehicle()`	Definer	Start-of-shift vehicle allocation.
`cleanup_on_logout()`	Definer	Frees vehicle, clears duty.
`has_completed_shift_start_today()`	Definer	The duty gate.
`get_dispatcher_kpis()`	Definer	Board summary in one round trip.
`raise_dispatch_alerts()`	Invoker	Per-department alert fanout.
`count_unread_alerts()`	Invoker	Authoritative unread badge count.
`search_everything()`	Invoker	Cross-module search.
`rate_limit_hit()`	Definer	Per-user request counting.
`run_automations()`	Definer	Scheduler entry point.
`citizen_update_location()`	Definer	Public token-authenticated GPS post.
`require_role()`	Definer	The guard opening every privileged mutator.

F — Migrations

#	NAME	PURPOSE
0001	`init`	Every V4 entity translated to Postgres.
0002	`rls`	Deny-by-default, then a policy per table.
0003	`realtime_storage`	Publications and storage buckets.
0004	`functions`	The multi-step workflows.
0005	`roles_alignment`	Adds Responder and Staff.
0006	`seed_mutual_aid`	All 41 partners.
0007	`v4_alignment`	Operator attribution, admin config, endorsements, performance.
0008	`fix_geo_trigger`	Location-trigger fix.
0009	`reporting_lines`	The reporting hierarchy.
0010	`alerts_push_integrations`	Alerts, receipts, push tokens, integrations.
0011	`route_incident`	Routing without committing a unit.
0012	`app_settings_perf`	Tuning store and hot-path indexes.
0013	`ops_links_cctv`	Command-bar links, CCTV wiring.
0014	`hr_leave_upgrade`	Approval trail, entitlements, indexes.
0015	`leave_self_insert`	Members file their own leave, Pending only.
0016	`groq_integration`	Groq as primary AI provider.
0017	`department_branding`	Per-department identity.
0018	`security_hardening`	Policy tightening.
0019	`privileged_function_guards`	Role guards on every privileged RPC.
0020	`revoke_public_execute`	Makes 0019's revocations take effect.
0021	`reporting_duty_allocation`	Saved reports and duty allocation.
0022	`realtime_unread_alerts`	Live unread count both directions.
0023	`duty_posts`	The establishment a shift must fill.
0024	`field_operations`	Offline-safe writes, honest timestamps, panic, numbering.
0025	`automation_engine`	Rules, run log, idempotency ledger, dispatcher.
0026	`automation_schedule`	The `pg_cron` job.
0027	`global_search`	Trigram search across six tables.

#	NAME	PURPOSE
0028	`rate_limits`	Per-user request counters.

APPENDICES G–H

Environment and Glossary

G — Environment variables

VARIABLE	USED BY	EFFECT IF ABSENT
`NEXT_PUBLIC_SUPABASE_URL`	web	Console shows the "not configured" screen.
`NEXT_PUBLIC_SUPABASE_ANON_KEY`	web	As above.
`SUPABASE_SERVICE_ROLE_KEY`	web	Push fanout and camera import unavailable.
`NEXT_PUBLIC_APP_URL`	web	SMS share links fall back to the request host.
`EXPO_PUBLIC_SUPABASE_URL / _ANON_KEY`	mobile	App runs but cannot reach the server.
`EXPO_PUBLIC_APP_URL`	mobile	Criti unavailable on the phone.
`GROQ_API_KEY`	web	Criti falls back to Gemini.
`GEMINI_API_KEY`	web	With no Groq key either, Criti explains it is unconfigured.
`BULKSMS_TOKEN_ID / _SECRET`	web	Citizen GPS request unavailable.
`REALPTT_ACCOUNT / _PASSWORD`	web	PTT console reports not configured.
`TOMTOM_API_KEY`	web	Traffic and POI panels report not configured.
`FLIGHTRADAR_API_KEY`	web	Flight feeds unavailable.
`MARINESIA_API_KEY`	web	Vessel feed unavailable.
`ITRAFFIC_API_KEY`	web	Camera import unavailable.
`OPENWEATHER_API_KEY`	web	Wind widget unavailable.
`HIKVISION_APP_KEY / _SECRET`	web	CCTV stays on the SANRAL registry.
`GOOGLE_SERVICES_JSON`	mobile build	Android build refuses unless `CRITICOM_ALLOW_NO_PUSH=1`.
`SUPABASE_DB_URL`	scripts	Migration runner and seeds cannot connect.

H — Glossary

TERM	MEANING
Silo	A department, and the isolation boundary around its data.
RLS	Row-level security. Postgres deciding, per row, who may read or write it.
Route matrix	`ROUTE_RULES` — which roles may open which paths.
Two-desk split	Call-takers log; dispatchers dispatch. Deliberately separate.
Routing	Telling departments a call is theirs. Distinct from assigning.
Assigning	Committing a specific appliance to a call.
Outbox	The responder app's durable queue of undelivered work.
Sync tray	The screen showing what is undelivered and why.
Claim	An automation's once-only record that it has acted on a thing at a tier.
Presentation	An alert's delivery style: `toast` or blocking `modal`.
Endorsement	A duty annotation — court, special event, instruction, restriction.
Establishment	The set of posts a shift must fill.
Publish / acknowledge	Making an allocation visible; the member confirming they have seen it.
PIR	Post-incident review, Fire & Rescue.
PCR	Patient care record, EMS.
SIRA	Private Security Industry Regulatory Authority — governs the guard reports.
OTA	Over-the-air update. JavaScript only; never native code.
Runtime version	The native-compatibility tag matching an OTA update to binaries that can run it.

Criticom V5 — Operations & Engineering Manual

Generated 4 August 2026 from the repository at commit `20ea2ec`. Every route, action, field name, table, procedure and figure in this manual was read from the codebase and the live database rather than recalled. Screen procedures follow the fields the code actually reads; where a form's fields are defined by a schema rather than named individually, the chapter says so.